



ACI desktop user interface manual

BASE24-eps[®]

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About this document

This manual provides the following information about the ACI desktop: framework, user security, user auditing, version checker, and online help.

The ACI desktop is the user access to the following ACI applications for configuration, operation, and management functions:

- BASE24-eps[®]
- Enhanced authorization for BASE24 applications
- Transaction Security Services (TSS)
- ACI Payments Manager
- ACI Smart Chip Manager
- ACI Card Management System
- ACI Token Manager

Audience

This manual is written for all users of the ACI desktop, primarily system administration personnel.

Prerequisites

Readers must know basic graphical user interface (GUI) functionality and procedures, such as using hyperlinks and buttons. The ACI desktop is organized by tasks (i.e., configure, system operations, and customer management) that require basic understanding of managing payment systems and applications.

Referenced documents

The **BASE24-eps Java server reference guide** provides detailed information on the data sources and facilities used to configure remote destinations for Java processes.

Software

This manual documents standard processing as of its publication date. Software that is not current and custom software modifications (CSMs) may result in processing that differs from the material presented in this manual.

Manual summary

Section 1, “Getting started,” provides an overview of the ACI desktop, windows, communication security, and information about default users. Also included is information about the user interface framework and user security.

Section 2, “Basic functions,” describes how to log in and log out and perform other functions available from the File menu.

Section 3, “User configuration,” provides an overview of user security, distributed security administration (DSA), and how to configure a user of the ACI desktop.

Section 4, “User management,” describes how to manage and monitor users of the ACI desktop. Includes information about user status and user permission information available on the View menu.

Section 5, “User auditing,” provides an overview of user auditing of the ACI desktop, including generating audit reports.

Section 6, “Version Checker,” describes how the Version Checker manages and updates ACI desktop versions.

Section 7, “Online help,” describes the desktop online help (JavaHelp) and how to use online help. Provides information about localizing the ACI desktop and customizing online help.

Section 8, “BASE24-eps window (UI) information,” information has moved to the BASE24-eps Windows Reference Guide.

Section 9, “Desktop.cfg file,” provides key-value pair information about the Desktop.cfg file.

Section 10, “SSL Keystore Password Manager,” provides information about using the Password Manager to manage SSL keystore passwords.

Section 11, “Multibyte character data support,” provides information about the BASE24-eps Multibyte Character Set Service module and the multibyte data fields that are available on ACI desktop windows.

Section 12, “Open source software license agreements,” provides OSS license agreements applicable to the ACI desktop.

Publication identification

The following information appears at the bottom of each page to identify this manual:

- Release and version number (R1.0v11.1)
- Publication date (Mar-2012)

Conventions used in this document



This manual uses the symbol shown at the left to alert to cross-industry best practices information for protecting sensitive data that is stored, processed, or transmitted in ACI applications. Carefully review such information to implement compliance to cross-industry best practices for securing data.

Refer to the product-specific Security best practices manual for more information about configuring an ACI product to comply with cross-industry best practices.

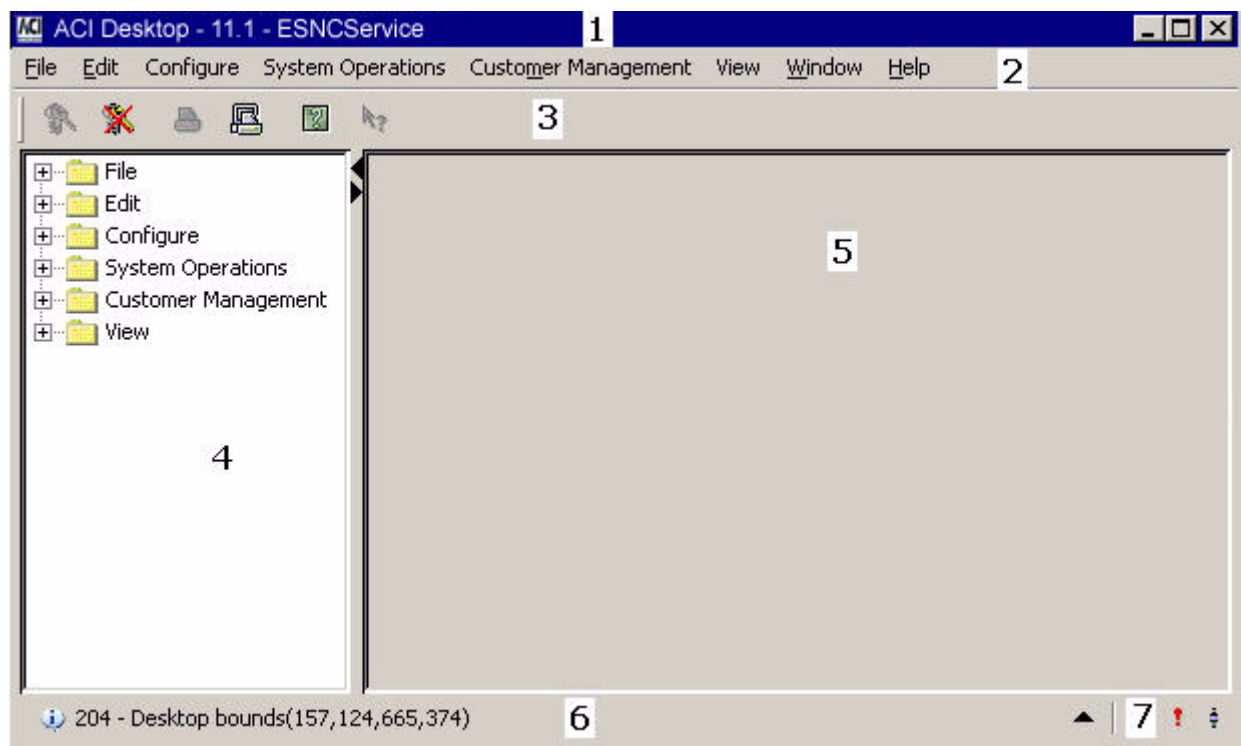
What's new

The following table highlights the major changes that have been made in the most recent update to the ***ACI desktop user interface manual***. The first column of the table lists the sections in which major changes have been made. The second column of the table describes the major changes for each section. These changes update the manual to release 1.0, version 11.1.

Section	Major Changes
All	Applies a new standard look and feel throughout the guide. All appendices have been converted to sections and section numbers are no longer provided in page numbering.
2	Adds a new topic describing how to change the look and feel of the ACI Desktop user interface client theme to that of a previous Windows version. When the ACI Desktop user interface client is installed on a Microsoft Windows 7 or Windows Vista platform, it has a different look and feel from previous versions.
12	Updates the Open Source Software License Agreements with current products and license agreements.

1: Getting started

The user interface of the ACI desktop consists of distinctive areas identified below.



- 1 - The **Title Bar** displays the name of the desktop.
- 2 - The **Menu Bar** provides access to available functions and application windows.
- 3 - The **Toolbar** contains buttons for common functions, such as log in, log out, display help topics and display context help.
- 4 - The **Navigation Tree** provides a hierarchical view of the menu bar.
- 5 - The **Application Area** contains opened application windows. Note that the text "LOGGING SENSITIVE DATA" is displayed if the debugging option on the User Preferences window is set, or if tracing of client messages exchanged with the server has been activated by a system administrator.

- 6 - The **Status Pane** contains informational messages about the activity between the desktop and servers. Right click on this pane to see the full detail of a message. Left click to see a list of previously received messages.
- 7 - The **Icon Area** provides informational icons. Place the cursor over an icon to display a popup message.

The key component of the ACI desktop is the **user interface framework**, which is a collection of Java classes, interface applications, rules, guidelines, recommendations, and styles that are used as a template for building and supporting ACI desktop applications. The components of the user interface framework include:

- User interface clients for which common functions, such as logging on and off the desktop, are available through the File menu
- Application Servers, such as Apache Tomcat or IBM WebSphere Application Server
Each application server is responsible for managing communication among desktop clients. Multiple application servers can be available in groups and are configured for the desktop using the local.cfg file. By defining groups, users can select application servers as applicable.
- Version Checker for automatic client version checking and updating
- Online help that is context sensitive

These components communicate with each other and with applications running on the application host server using a variety technologies. XML messages are passed to application servers on the host using Transmission Control Protocol/Internet Protocol (TCP/IP).

Hypertext Transfer Protocol Secure Sockets Layer (HTTPS) is used for ACI desktops to communicate with web servers clients.

Communication security

Connections among ACI desktops, web servers, and application servers on the host can be configured to be secured using Secure Sockets Layer (SSL).

Certificates are used to authenticate servers to desktop clients, and optionally authenticate clients to servers. Public key cryptography is used to encrypt messages sent among clients and servers.

Note: SSL is configured using SSL specific properties defined using the Application Configuration window (**System Operations > Server Management > Application**). Refer to the **BASE24-eps Java server reference guide** for more information.

The following Java™ Archive files provide the support needed for SSL:

- Jcert.jar

- Jnet.jar
- Jsse.jar
- uif_sslkeys.jar, which contains the default keystores.

Certificates and private keys used for encryption and authentication are maintained in protected files called keystores. These files vary by application and are identified by .sto.



Note: Custom communication developed and implemented for TCP/IP lines outside of the ACI desktop must be secured using SSL.

User security

User security (USEC server) consists of the authentication of users, authorization of user activity, management of active user sessions, and the logging of user activity. The user.security.version property provides the USEC server version.

For authentication, an individual must provide a user ID and password. This information is checked the user security database to determine whether the individual can access the application. This authentication includes the requirement that the user ID be defined in the user security database and the password must match that recorded for the ID. In addition, the user's security group and password group restrictions are checked.

User authorization is the validating of each activity initiated by a user. During user authorization the activity initiated by a user is compared to the user's security profile, which defines the functions (i.e., Delete, Save, and View) that a user can perform. A user can only perform those functions that have been assigned to the user's profile. This ensures that a user does not perform an activity not authorized by the user's security configuration.

See section 3 (User configuration), section 4 (User management), and section 5 (User auditing) for more information.

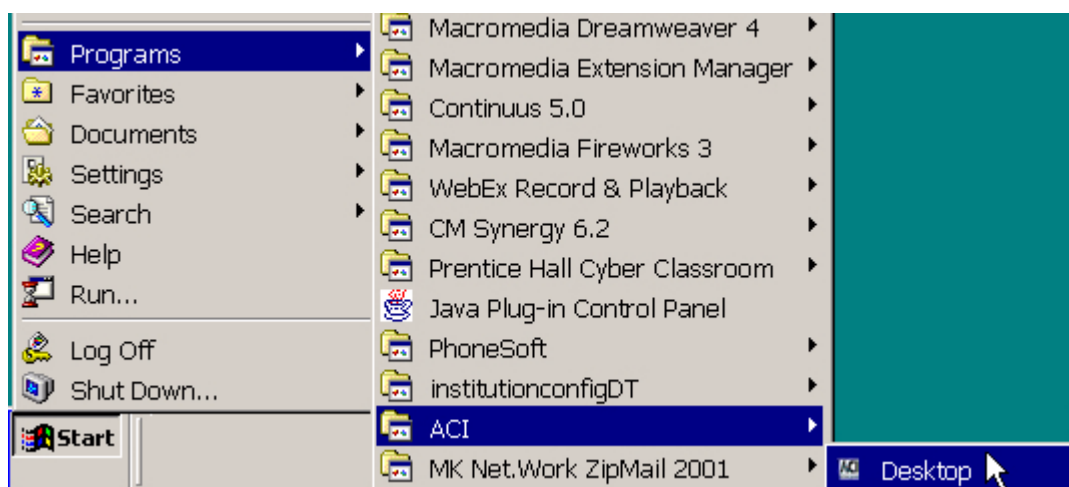
ACI desktop modules list

A modules.lst (modules list) is included with the install of the ACI desktop. This modules list is required because it contains module identifiers related to user interface windows, or more specifically, ACI desktop functionality. If additional functionality is added to the desktop after an install, the modules.lst file must be updated to include the new module ID(s) pertaining to the new functionality, and the desktop must be restarted to activate the new functionality.

On all platforms, the modules.lst file is located in the <eshome./ESWeb/ESConfig folder.

2: Basic functions

Each desktop session is started by running the **RunDesktop.bat** batch file, which is usually configured during installation to be accessed through a shortcut located on a client desktop or a menu. For example:



After the desktop establishes contact with the applicable servers and JAR files have been downloaded (see the next page for more information), the desktop displays the Log In window for entry of a user ID (name) and password. The desktop opens after a valid user ID (name) and password combination has been provided and authenticated. The file menu functions (e.g., Configure) and windows that are presented are dependent upon the user's configuration settings.

This section discusses logging on to the desktop, connecting to application servers, setting user preferences, and viewing version information.

Enable SSL for the BASE24-eps

An SSL enabled desktop has the  lock image displayed at the lower right. Use the following procedure to enable SSL for a BASE24-eps desktop.

1. Access and open the desktop.cfg (local.cfg) file.
2. Add the following line to the [Desktop] section:

```
SSLEnabled=true
```

For example:

```
[Desktop]
.....
SSLEnabled=true
Host=TomcatServerName
.....
```

3. Add the https port entry in the hosts section, setting the port to the value configured in the server's application server configuration file. In a default deployment this is the Tomcat server.xml file. (The servlet name and port may vary by product release; match the one used in your vanilla deployment).

```
[Hosts]

TomcatServerName=<tomcat_machine_name>:8443/ESWeb/ESHTTPServlet
```

4. Add the following SSL section if not already present

```
[SSL]
#
# The provider and protocol handler package need to be
# specified unless otherwise defined in the java.security
# file. They should be set to sun's implementation(s).
# Provider=com.sun.net.ssl.internal.ssl.Provider
# ProtocolHandlerPkgs=com.sun.net.ssl.internal.www.protocol
#
# If auto update is true, the specified key store file below
# will automatically be extracted by locating it in any jar
# in the classpath. This typically would come from the
# uif_sslkeys.jar. If the keystore should not be extracted
# upon startup, then set thisproperty to false.
#
KeyStoreAutoUpdate=true
KeyStorePassword=defaultDT
KeyStore=dtKS.sto
KeyStoreType=JKS
```

5. Import the server's certificate into the keystore.
6. Restart the Desktop. You should have:
 - ☐ No URL connections errors in the desktop.err file. This means that there was a HTTP SSL connection established between the desktop and Tomcat.
 - ☐ You should see a lock icon on the right hand bottom corner of the desktop application frame. This confirms that the desktop is started with SSLEnabled=true
 - ☐ Logon to the desktop and test.

Additional information on the RunDesktop.bat file

The install program names the RunDesktop.bat file. When the RunDesktop.bat is run, it starts up a JVM and launches the VersionChecker Client program which determines whether new and updated JAR files are available from the Server Repository (see section 6 (Version

Checker) for more information). When the Version Checker is completed, the batch file calls the DesktopLauncher.exe that starts up the Desktop. The DesktopLauncher.exe also accesses a file named DesktopLauncher.lax.

For support purposes, ACI does not recommend that the above files be renamed. However, if it is desired to rename the files, the following considerations apply:

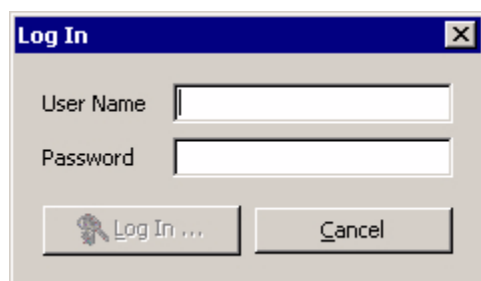
- The RunDesktop.bat file can be renamed after install. Any shortcuts pointing to this file will need to be modified to reflect the name change.
- If the DesktopLauncher.exe file is renamed, the RunDesktop.bat file will need to be modified to reflect the name change. Also, the DesktopLauncher.lax will need to be renamed to the same name as the new name for DesktopLauncher.exe.

Connection information

This topic provides information for logging on, concurrent sessions, connecting to an application server, changing passwords, and logging off.

Log in (log on)

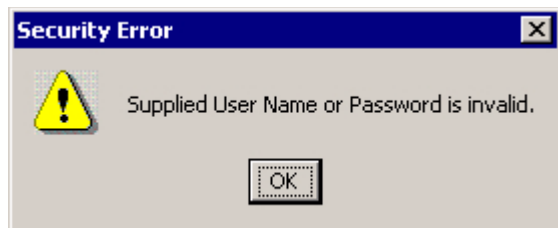
To log in, select the Log In function located on the File menu (**File > Log In**), the Log In button () on the toolbar, or navigation tree.



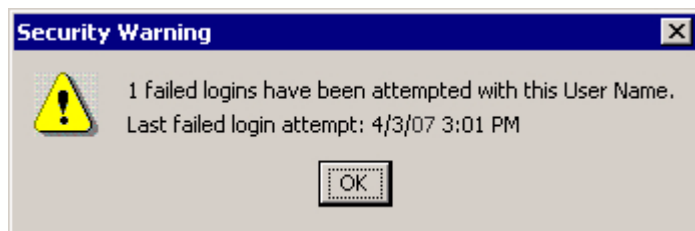
After a successful log-on attempt (authenticated user), the desktop provides access to those applications (windows) that system administration has assigned to the user (i.e., the permissions and permission groups of the user's profile). Each session is assigned a unique session number used for authorization purposes.

In addition to the above, session and window (UI) timeout values are initiated and tracked during a session. A timeout is a defined period of inactivity. Timeout limits are defined in the security groups assigned to users.

If the user name and password are not correct or the user name has been disabled or does not have valid access at the time of login, the user is not given access to the desktop. In this case a security error message (shown below) is displayed. The error message is purposely vague so not to indicate which information (name or password or both) is incorrect.



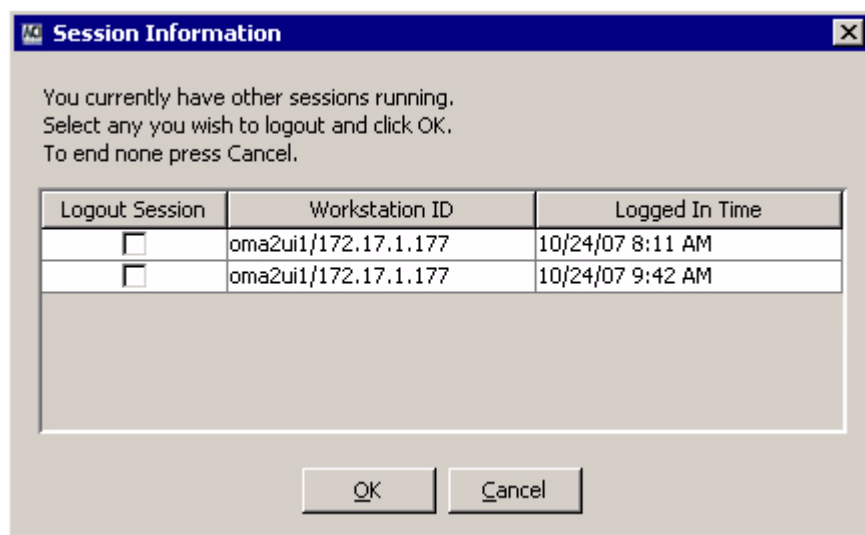
A **security warning**, such as the example below, is displayed upon a successful log-on attempt if there were one or more unsuccessful log-on attempts before the user successfully logged on.



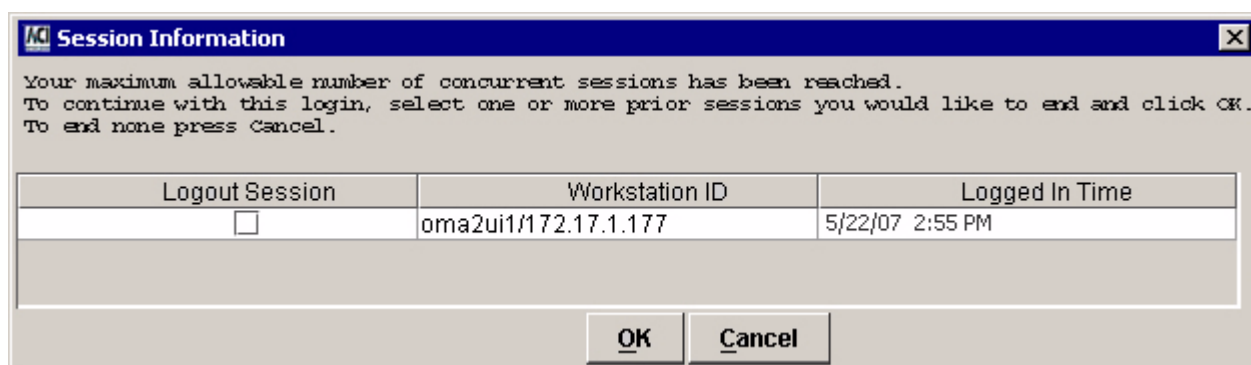
Concurrent sessions

The number of **concurrent active sessions** allowed for a user is limited to a maximum as defined in the security group of the user's profile. A possible scenario in which a user might have concurrent active sessions is by logging on from another terminal without logging off from an active session that was started at another terminal, or in a browser session.

This version of the Session Information dialog is displayed if a user has one or more existing active sessions when logging on, and this number of sessions is fewer than the maximum number as defined in the user's security group.



This version of the Session Information dialog is displayed if a user, upon startup, has already reached the maximum number of allowed active sessions defined in the user's security group.



Connect to an application server

During startup, the ACI desktop attempts to connect to the default (or most recently used) application server. If multiple application servers are available, a user can access the Connection window (**File > Connection**) to select an alternate. The Connection window displays a selectable list of available application servers from which one can be selected.

If the one selected accesses a different shared user security database, any existing active sessions for the user are terminated. Also, the port used must be available (i.e., not used by another system or application); otherwise, a connection error is encountered.

A system administrator can configure the list of supported hosts using the host variable in the desktop.cfg file. See section 9 (Desktop.cfg file).

Display name of the user that is logged on to the desktop

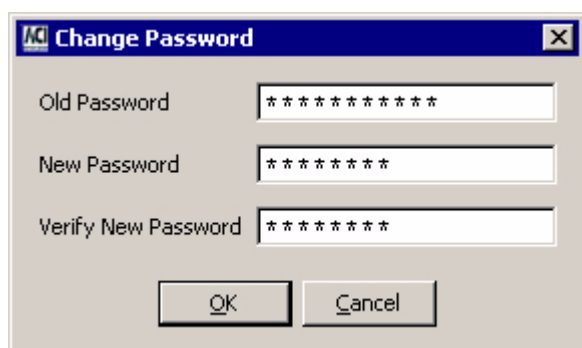
To obtain the name of the user currently logged on to the ACI desktop, mouse over the user icon () that is located at the bottom right of the desktop. For example:



If this icon is not displayed, access the desktop.cfg file and set the **ShowLoggedOnUser** property, which is located in the [desktop] section of the file, to true. This property can be added if it is not present. The desktop will need to be restarted if a change is made to the desktop.cfg file.

Change a password

To change a password, log in to the desktop and select the Change Password function (**File > Change Password**) to display the Change Password window:



Note: This dialog box will not be available when you are using a third-party authenticator other than the one provided with this application. Contact your security administrator for details.

Certain password restrictions and requirements apply. See section 3 (User configuration) for more information.

Log out (log off) and exit

To log out, select the Log Out function located either on the File menu (**File > Log Out**), the Log Out button () on the toolbar, or navigation tree.

Logging out does not close the desktop. To close the desktop, select **File > Exit**.

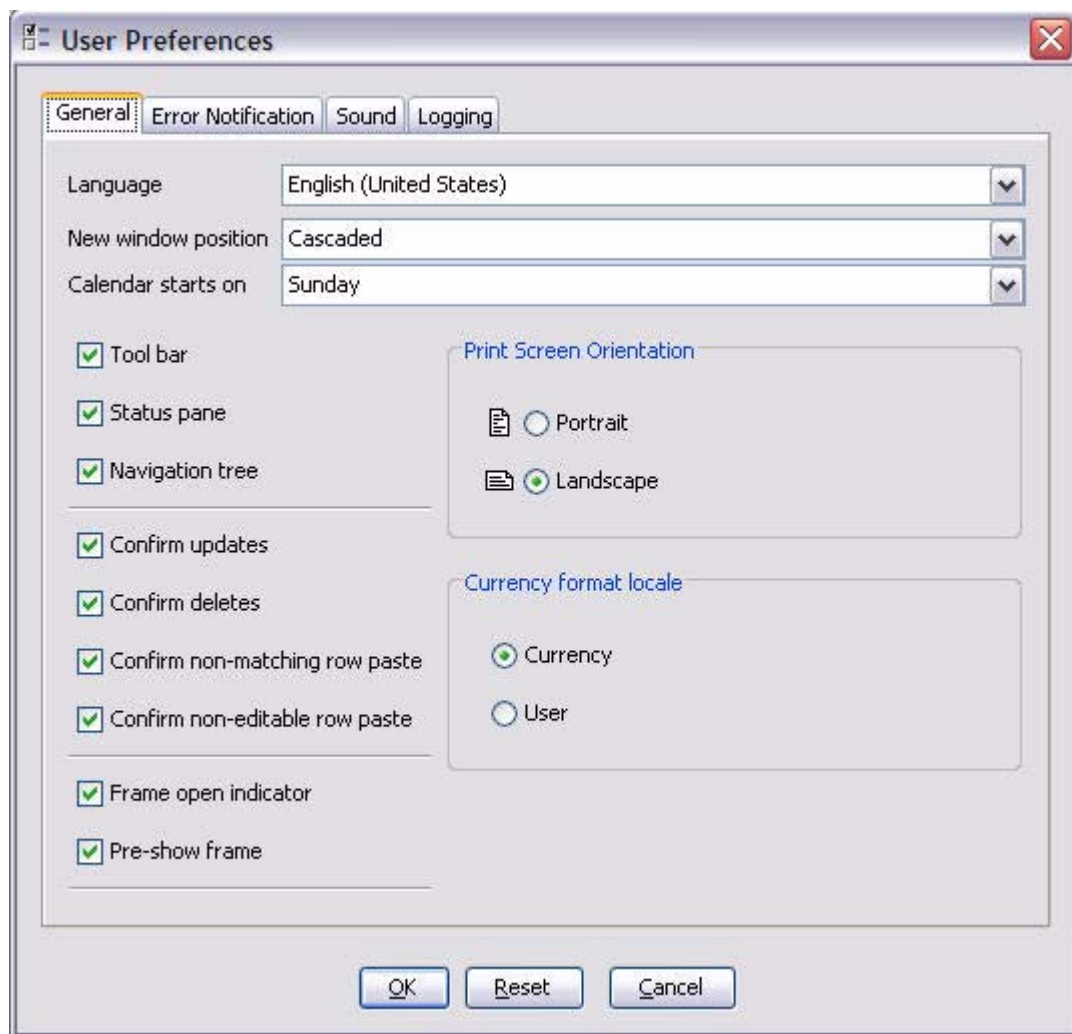
Note: A user does not need to be logged in to exit. Also, if a user exits without logging out, the user is automatically logged out from the desktop.

User preferences

Each user can set desktop characteristics to desired preferences by accessing **File > Preferences**. These preferences are discussed below.

General preferences

Use the General tab of the User Preferences window to set the following user preferences.



Descriptions of general preferences are discussed below.

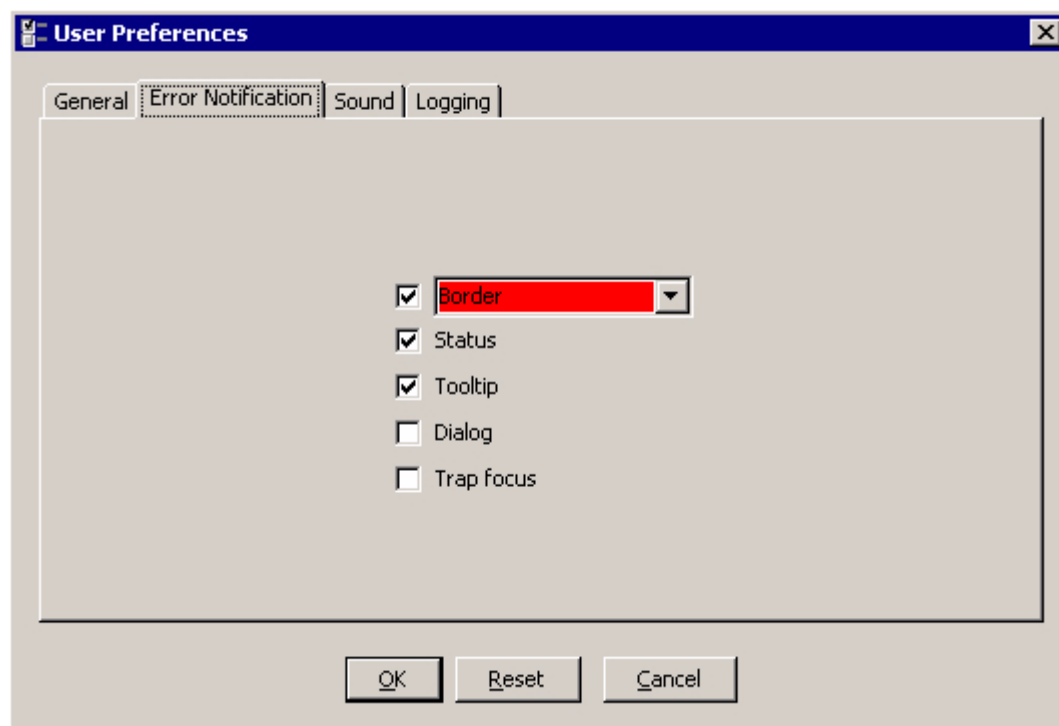
Preference	Description
Calendar starts on	Indicates what day of the week calendars start on in the windows. For example, if this is set to Wednesday, all of the windows that have calendars will start on Wednesday, rather than the standard Sunday.
Confirm deletes	Indicates whether the desktop displays a confirmation dialog box before a delete function is processed.

Preference	Description
Confirm non-editable row paste	Displays a dialog box indicating that an attempt was made to paste a copied table row onto a non-editable row.
Confirm non-matching row paste	Displays a dialog box indicating that an attempt was made to paste a copied table row onto a table row that has a different number of table cells.
Confirm updates	Indicates whether the desktop displays a confirmation dialog box that requires a response before an update function is processed.
Currency format locale	Sets whether the default currency is obtained from the desktop configuration or user configuration settings.
Frame open indicator	Indicates whether a frame open indicator (which is the phrase "Opening window...") is displayed in the desktop toolbar while a window is opening.
Language	Displays menu functions, dialog box text, messages, and field names in the selected language, if the selected language is defined in the properties file for a window. This also allows access to available localized online help (online help provided in a different language or dialect). Note: If the locale property is not set, the language is the default locale of the JRE (Java Runtime Environment) installed on the client's workstation.
Navigation tree	Sets whether the navigation tree is displayed upon desktop startup. If checked, the navigation tree is initially displayed. If not checked, it is initially hidden.

Preference	Description
New window position	Sets the position of a window when opened as follows: • Bottom • Cascaded • Centered • Frame specified • Left • Right • Top • Top left
Pre-show frame	Indicates whether a blank window is shown while a window is initializing (being opened).
Print screen orientation	Sets the default print orientation to either portrait or landscape.
Status pane	Indicates whether the status pane is displayed.
Toolbar	Indicates whether the toolbar is displayed.

Error notification preferences

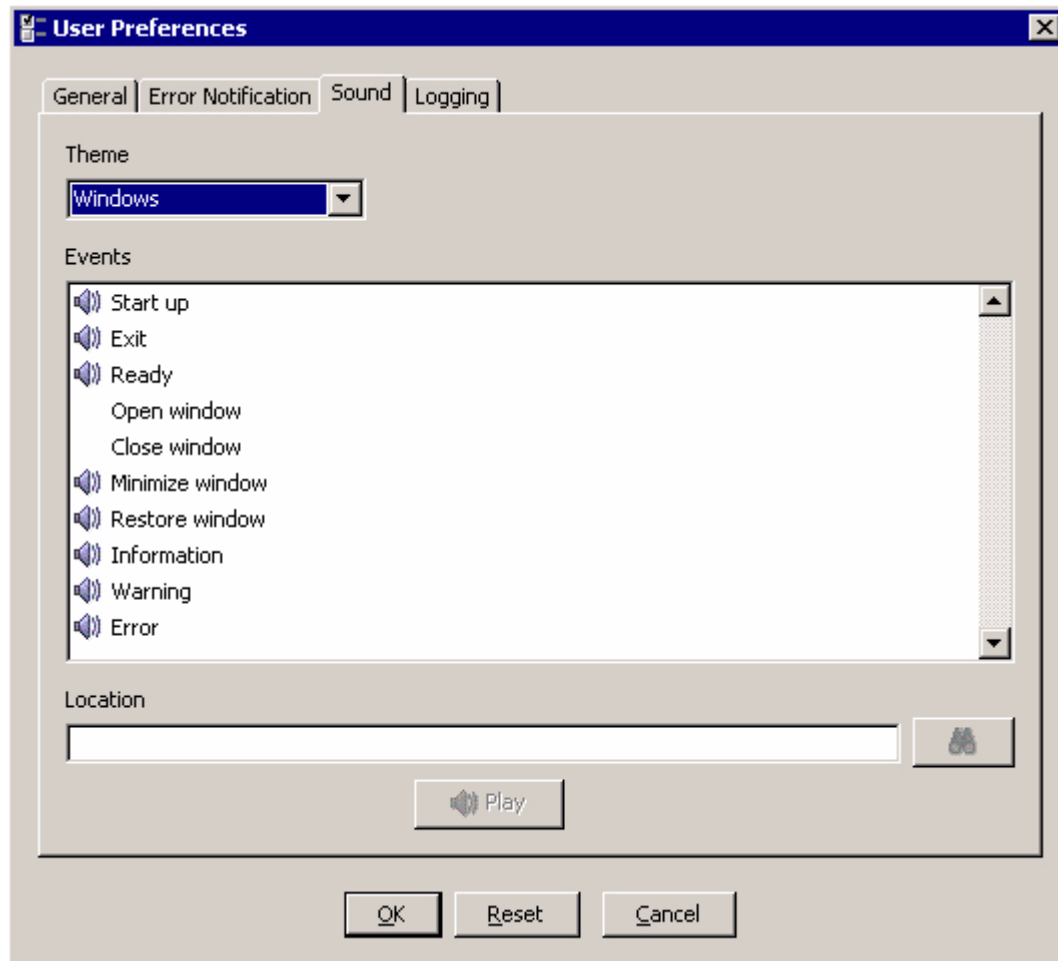
Use the Error Notification tab to set error notification preferences.



Preference	Description
Border Color	Sets the border color of fields that are in error.
Status	Indicates whether error text is displayed in the status pane of windows.
Tooltip	Indicates whether error messages are displayed in tooltip text. Tooltip text is text that appears in a pop-up window over a field or button.
Dialog	Indicates whether error messages are displayed in a dialog box.
Trap Focus	Makes error messages modal; that is, a user is required to resolve an error before focus can be changed to another window or tab.

Sound preferences

Use the Sound tab to set the sound played during certain events (as selected in the Events field).

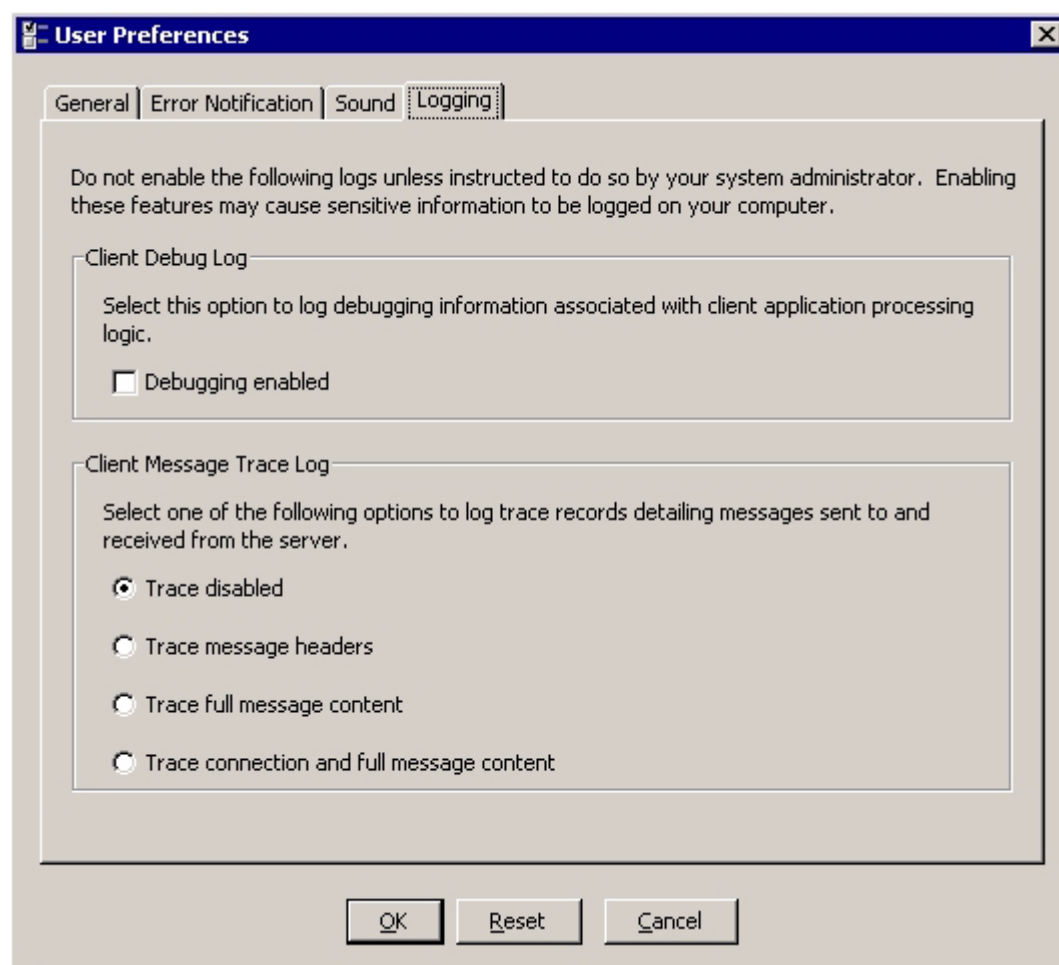


Preference	Description
Themes	Sets the sound group to apply to the selected event(s). Themes are defined in the sound.cfg configuration file located in the desktop.jar. The theme selected is stored in the configuration file of the logged on user.
Events	An event to be associated with a sound file.
Location	The location of the sound file, such as a .AU file (Java) and .WAV file (Windows), to associate with the selected event.

Logging (debug and trace) preferences



Per PCI requirements, the ACI desktop will not log sensitive data to a client's computer unless configured to do so using the Logging tab of the User Preferences window.



Though the User Preferences window and the Logging tab are available from the File menu and can be altered before a user logs on to the ACI desktop, the changes are only activated after the user logs on to the desktop and the user is determined to have the permission to make the change. If the user does not have permission to set debugging and trace levels, the change is not made.

Debugging



When the Debugging enabled option is selected, debug messages are written to DeskTop.dbg debug file and the text "LOGGING SENSITIVE DATA" is displayed on the background of the application area.

Note: An ACI desktop user can only enable debugging if the view function of the **allowPostLogonDebug** task ID has been associated with the user's profile.

Trace

Three options are provided on the Logging tab for enabling tracing:

- Trace message headers
- Trace full message content
- Trace connection and full message content



If one of the above options are selected, trace messages written to the DeskTop.log file and the text "LOGGING SENSITIVE DATA" is displayed on the background of the application area. By default and per settings in the LogScrubber.cfg file, the following information is scrubbed (i.e., information is masked with asterisk (*) characters):

- Passwords
- Client IP address
- Session information

Note: An ACI desktop user can only set the tracing level if the view function of the **allowPostLogonTrace** task ID has been associated with the a user's profile.

Look and feel impacts for Microsoft Windows Vista and Windows 7

When the ACI Desktop user interface client is installed on a Microsoft Windows 7 or Windows Vista platform, it has a different look and feel from previous versions.

If desired, you can change the look and feel theme to that of a previous Windows version and to what is used in the BASE24-eps documentation. Use the following steps to change the Windows control panel theme setting.

1. Logon to the Windows PC.
2. Go to Start - Control Panel.
3. Navigate to Appearance and Personalization.
4. Click on Change the theme.
5. Scroll down to Basic and Contrast Themes.
6. Select "Windows Classic". The "Windows Classic" theme provides the look and feel that was available in previous Windows versions and pre-11.1 BASE24-eps ACI Desktop user interface clients.

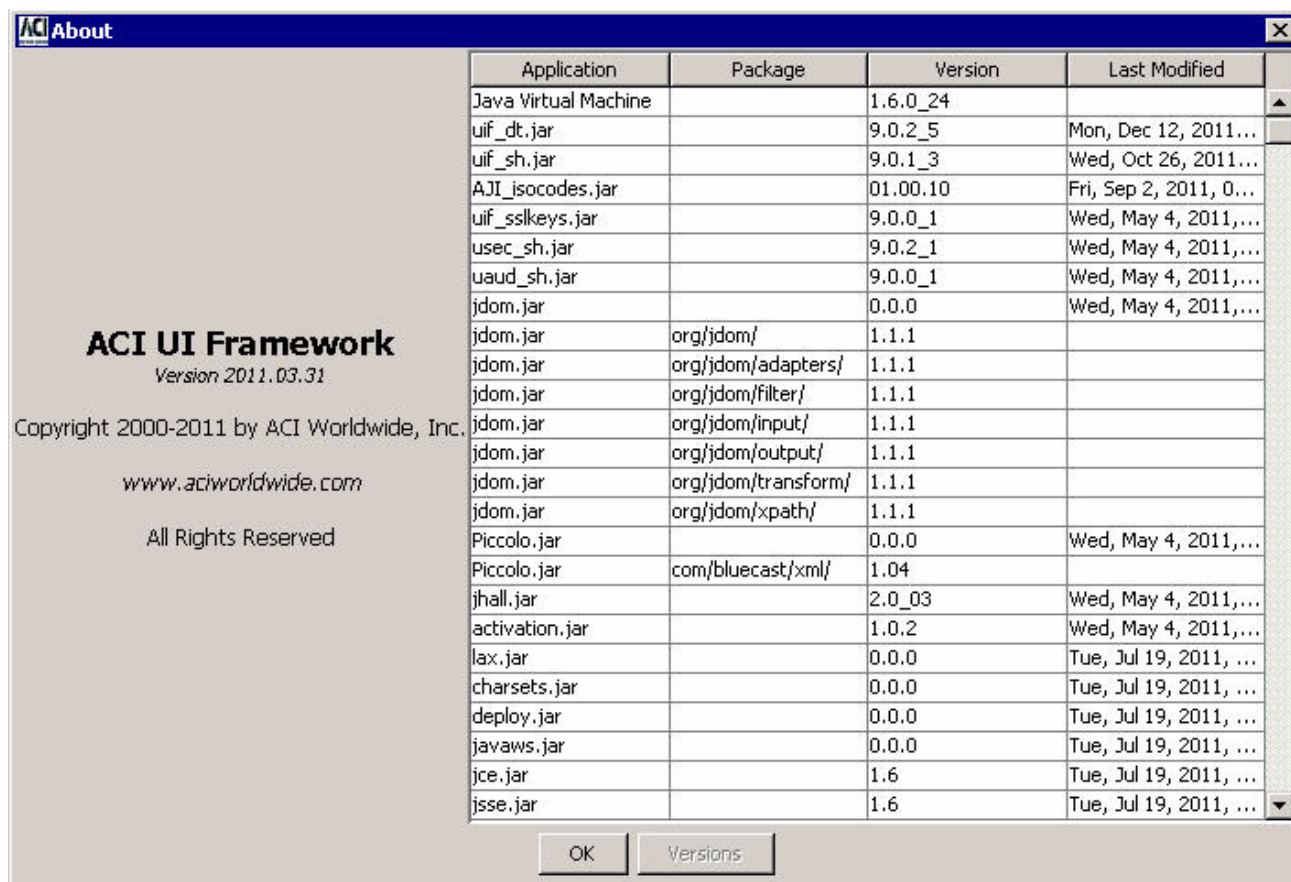
7. Exit the Control Panel.

Display Framework version information

Access **Help > About** to display desktop version information as shown below:

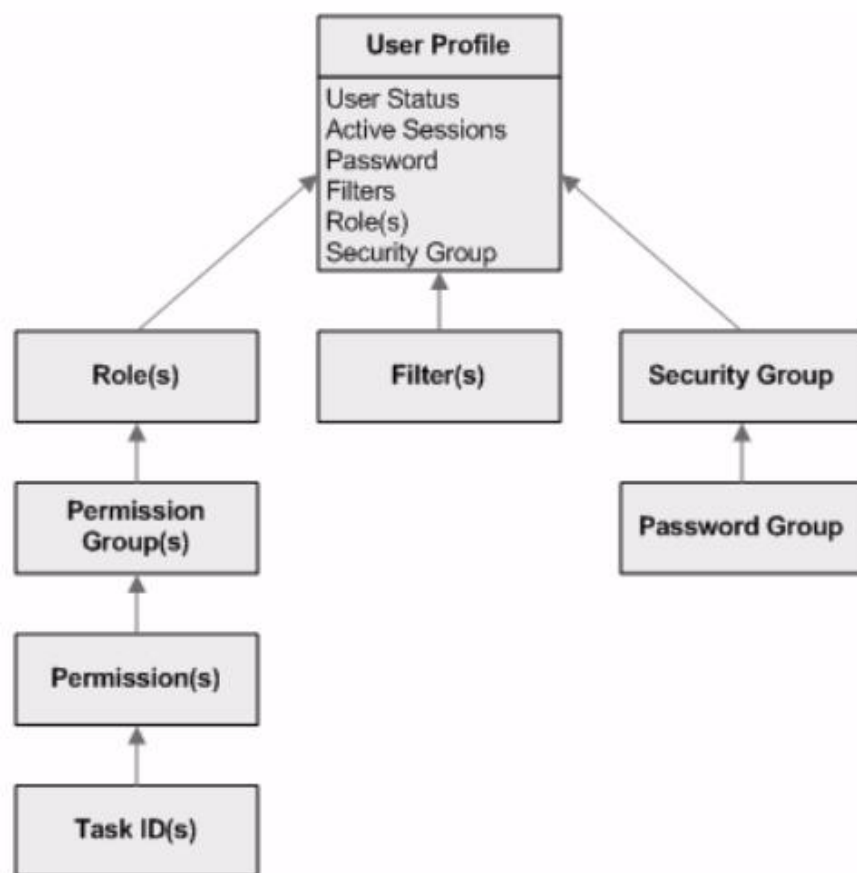


Click **Versions** to display detailed version information as shown below:



3: User configuration

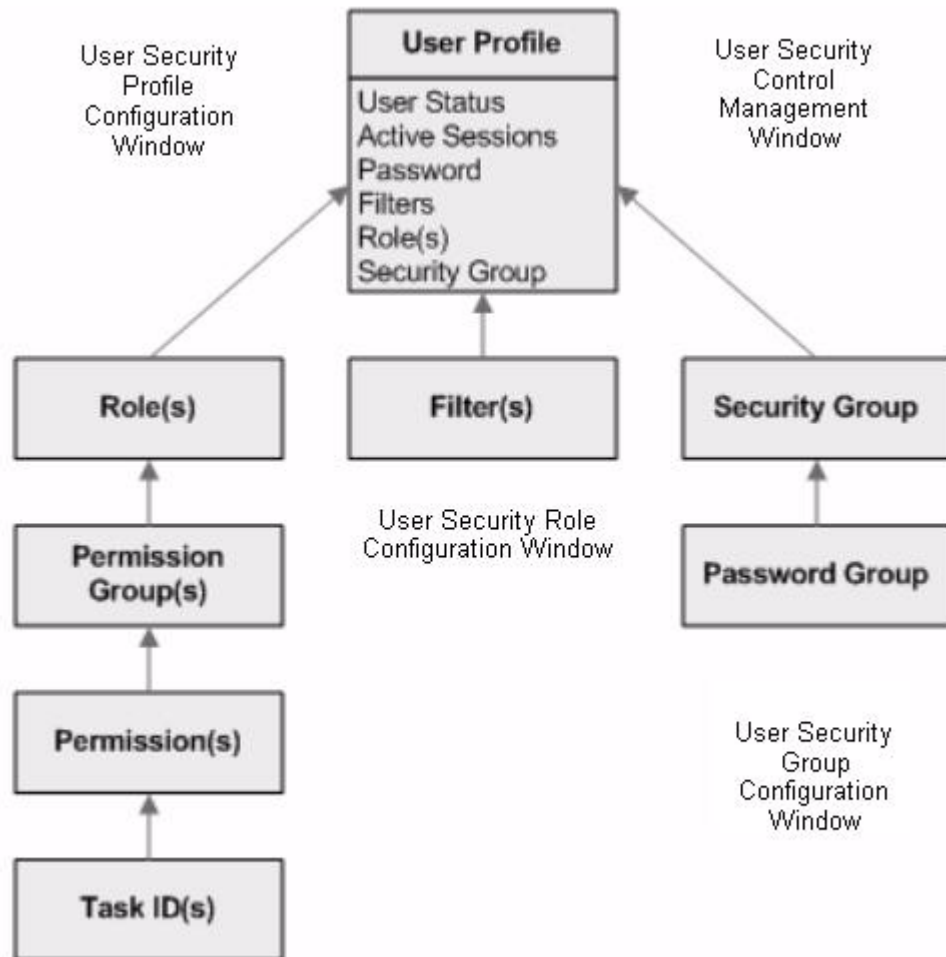
The ACI desktop provides access to an integrated user security environment in which each user must be defined in a user security database. This section consists of two parts: an overview of user security and steps about how to add a user. It also discusses the user profile which is central to establishing a user's access level to the ACI desktop. Each profile defines the integral components of user security, some of which, such as security and password groups, can be shared among multiple users. The hierarchy of a user profile is shown below.



Note: The user.security.version property in the APPLCNFG table is configured to use the USEC 1.2 mode (the BASE24-eps default). This mode enables filters to be tied directly to user profiles, thereby improving the performance of user security.

User Security Configuration window usage

User profiles are maintained from the ACI desktop using various user security windows. The hierarchy of a user profile from the previous page is shown below with the names and usage of the user security windows added.



All user security windows are accessed at **Configure > User Security**. A comprehensive hierarchical view of the relationships among tasks IDs, permissions, permission groups, and roles can be accessed at **View > User Permissions**.

User profiles

Each user is assigned a **user profile** consisting of the following configuration components:

- User ID — A unique character string used to identify a user.

- General user information — Name and contact information about a user.
- Filters — Filter IDs directly assigned to the user.
- User status — In response to certain events, such as too many bad log-on attempts, a user's status can be changed (i.e. disabled) programmatically or by security administrative personnel.
- Role(s) — One or more sets of permission groups.
- Security group — A unique ID that defines a set of common user security access restrictions.
- Password group — A unique ID that defines a set of common user password restrictions.

Roles

A role defines the level of a user's access to the desktop. Each role identifies a job function, such as security administrator, system administrator, manager, and end user. Roles apply a set of permissions, tasks, and functions to users.

A role can contain one or more permission groups. A permission group can contain one or more permissions. Each permission is associated with a task (window application) and defines the functions (e.g., delete, save, send, and view) allowed for that task.

Permission groups

A permission group is a collection of selected permissions that is assigned a unique identifier (the permission group ID) and a description of the purpose or use of the group.

Permissions

Permissions define specific access to a window. Each permission consists of a single task (task ID), the functions that can be performed within the task, a unique ID (the permission ID), and a description of the purpose or use of the permission.

Field filters

Field filters restrict a user's access to field values by defining either a single value, multiple values, or a range of values allowed.

Note: The filter IDs available vary by application. However, a common example is to restrict a user to see only data for a single institution.

Task IDs

Each task consists of a unique identifier and a group of functions (e.g., delete, save, send, and view) and corresponds to a window on the ACI desktop. Available task IDs and functions for each task ID are provided on the Permissions tab of the User Security Role Configuration window and are used to create permission IDs. Task IDs can be used in multiple permissions.

Passwords and password groups



Beginning with version 08.2, a user's password is stored in the user security database in encrypted form using a Secure Hash Algorithm with Randomized Salt Hash (SHA-256) that is not reversible. This ensures that passwords cannot be extracted in clear form from the database. Only SHA-1 is supported for 07.4 and earlier versions. SHA-1 is also supported in version 08.2 for backward compatibility.

Note: When a new user is added, the option to force the new user to change his or her password on first use can be selected as described in step 4.4 later in this section.

Passwords comply with Payment Card Industry complex password requirements as follows:

- Passwords are case sensitive.
- Alphabetic, numeric, and special characters are permitted.
- Password history is maintained.
- Passwords cannot be shared.
- Characters cannot be repeated sequentially. For example, a password with the string "BB" appearing anywhere in the password is not valid.
- The password cannot be the same as the user ID. For example, a user with the ID "enduser02" cannot use "enduser02" as the password.
- A user's password must meet the requirements (discussed below) defined in the password group assigned to the user.

Password group requirements

Password groups consist of password requirements that are imposed on all user passwords assigned the group. A password group consists of the following password requirements as configured using the Password Groups tab on the User Security Group Configuration window.

- The number of days prior to a password expiration on which the desktop displays a warning dialog prompting a user to change the password.
- The maximum number of days a user can use a password.
- The minimum password length (4 characters or more).



Note: Payment card cross-industry best practices specify a minimum length of 7 or more characters.

- The maximum password length.
- The minimum number of numeric characters required in each password.
- The minimum number of alphabetic characters required in each password.
- The maximum number of old passwords retained for each user (users cannot use retained old passwords).
- The maximum number of password changes allowed during a 24-hour period.

Security groups

Each user must be assigned one and only one security group. However, since a security group can be assigned to multiple users, the security group simplifies the management of user security, assigns passwords by group, and assists in implementing distributed security administration, which is discussed later in this section.

Each user security group consists of the following common user security access restrictions that apply to all users assigned the security group:

- The time each day a user can log on to the desktop.
- The time each day a user must be logged off the desktop and, after which, can no longer log on to the desktop.
- The days of the week on which a user can log on to the desktop.
- The maximum number of invalid log-on attempts allowed for each user. If this value is exceeded, the user's status is disabled.

If the environment variable is false (the default), then the user must wait the required 300 seconds before they can once again attempt to log on with their user ID. If the environment variable is true, the system administrator must intervene and manually reset the user status to Active. Refer to the **BASE24-eps Java server reference guide** for more information about the `isDisableUserOnFailedLogonAttempts` property.

- The number of seconds, if set, after which the number of invalid log-on attempts is reset to zero.
- The maximum number of active sessions allowed for a user.

When a user fails to log off correctly from the desktop or if a system outage occurs during a user's active session, the user security database still maintains a record of the active session. Therefore, if the maximum number of active sessions has been reached by a user attempting to log on to the desktop, the user is presented with an option to stop one or more of the active sessions.

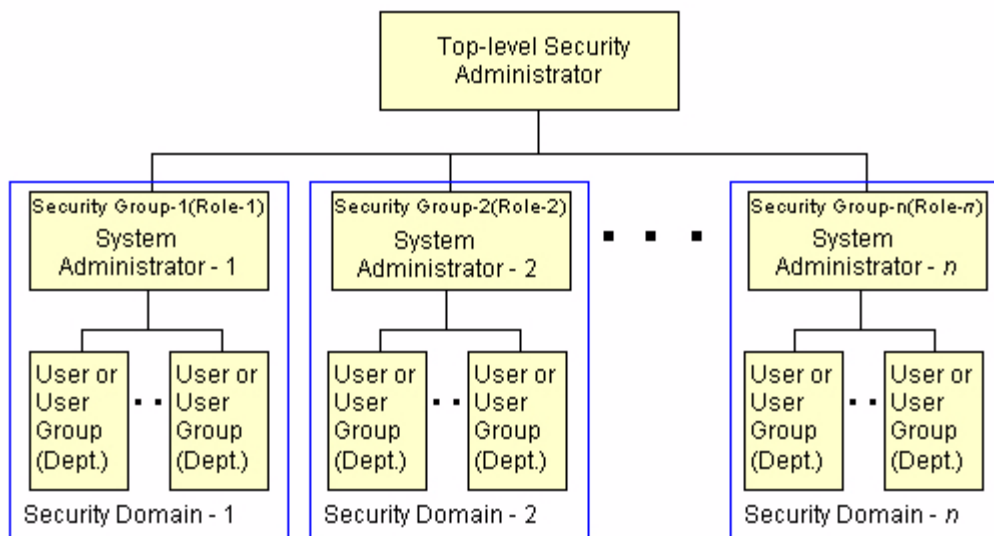
- The number of seconds of user inactivity after which the user is automatically logged off the desktop (this is called a **session timeout**).
- The number of seconds of terminal inactivity after which the desktop is disabled and any logged on user is logged off (this is called a **UI timeout**).
- The number of days a user's ID can be inactive (not used) after which the user ID status is disabled.
- The password group ID defining password requirements applied to all users of the security group.

Distributed Security Administration (DSA)

Each installation is configured with a system administrator and security administrator. The system administrator has access to all applications of the desktop. The security administrator has access to the user security windows. Distributed security administration involves the security administrator configuring other, lower-level, administrators that are assigned a limited set of permissions.

Distributed security is implemented by the top-level security administrator by configuring applicable roles, assigning these roles as necessary to security groups, and assigning the security groups to other, lower-level, system administrators.

The top-level security administrator can set up tiers of other system administrators who are limited to a **security domain**, which is a set containing a security group, password group, roles, and permissions as defined by a security role ID. At this level, a role ID is used to restrict users within the domain to a defined set of permissions (including permission groups), tasks and filters. This methodology can be used to create system administrator groups and departments as shown in the illustration below.



For example, the system administrator assigned Role1 can create user profiles or groups of user profiles having roles that can contain only those permissions, permission groups, tasks, and filters defined for Role1. With distributed security administration, users, including administrators, cannot access the users and roles of other security domains.

Steps to add a user

Note: Some steps have additional steps (i.e., steps 3.1 – 3.3 and 4.1 – 4.5). Whether these additional steps are required depends on the extent to which a complete user security system has been configured.

- Step 1.0: Configure a Password Group
- Step 2.0: Configure a Security Group
- Step 3.0: Configure a User Role
 - Step 3.1: Configure Field Filters
 - Step 3.2: Configure Permissions
 - Step 3.3: Configure Permission Groups
- Step 4.0: Configure the User's Profile
 - Step 4.1: Set the User's Status
 - Step 4.2: Assign a Security Group
 - Step 4.3: Enter Information about the User
 - Step 4.4: Set the User's Password
 - Step 4.5: Assign Roles
 - Step 4.6: Assign Filters

Usage Note: To ensure that changes are implemented in the user security database, click Save () each time a step has been completed.

Tip: If a new user is to be based on an existing user profile, that is, the password group, security group, and roles are already configured, go directly to step 4.0 and read the Tip provided in the step.

Step 1.0: Configure a password group

Use the Password Group tab on the User Security Group Configuration window to configure a password group.

The screenshot shows the 'User Security Group Configuration' window with the 'Password Group' tab selected. The window has a title bar with standard Windows controls. Below the title bar is a toolbar with icons for file operations. The main area contains two tabs: 'Security Group' and 'Password Group'. The 'Password Group' tab is active, showing a 'Password Group' dropdown menu and a 'Description' text box. Below these are two columns of settings, each with a label and a spin box:

Minimum Length	7	Lifetime Days	30
Maximum Length	10	Old Password Cache	5
Required Numeric Characters	2	Expire Pre-Warn Days	7
Required Alpha Characters	2	Changes Per Day	3

In the example above, the password group **SysAdminPswdGrp** is shown. This group is described as the default password group for system administrators. The password requirements are set in the fields provided on the tab.

Password groups are assigned to users (their profiles) through security groups. Therefore, a password group must be assigned to a security group using the Security Group tab of the User Security Group Configuration window. The security group is then used in user profiles.

Step 2.0: Configure a security group

Topics in this step to consider are listed below and are discussed on the following pages.

- UI and Session Timeout Considerations
- System Administration Role Filter (Owner ID)

Use the Security Group tab on the User Security Group Configuration window to configure a security group.

The screenshot shows the 'User Security Group Configuration' window with the 'Security Group' tab selected. The configuration is as follows:

- Security Group:** SysAdminSecurityGrp
- Description:** Default security params for system administrators..
- Password Group:** SysAdminPswdGrp
- SysAdmin Profile:** (empty)
- Allowed Time and Days:**
 - Begin:** 08:00 am
 - End:** 05:00 pm
 - Days:** Sun, Mon, Tue, Wed, Thu, Fri, Sat (all checked)
- Bad Logins Allowed:** 4
- Reset Bad Logins:** 300 seconds
- Sessions Allowed:** 5
- UI Timeout:** 1200 seconds
- Inactive Days Allowed:** 60
- Session Timeout:** 0 seconds

In the example above, the configuration for the **SysAdminSecurityGrp** security group is shown. This group defines standard authentication requirements for system administration user. The password group **SysAdminPswdGrp** is assigned to this security group. As a result, the password requirements defined by **SysAdminPswdGrp** apply to all users of this security group.

UI and session timeout considerations

A security group defines both UI and session timeouts. These timeouts are defined in units of seconds using the Security Group tab of the User Security Group Configuration window.

A UI timeout occurs after a duration of desktop inactivity defined in the UI Timeout field. When a UI timeout event occurs, the user no longer has access to the desktop and any open windows. However, the user's session and the open windows are not closed, and the desktop displays the Verify Password dialog in which the user must provide a valid password to restore access to the desktop and any open windows. If the user does not provide a valid password in three attempts, they will be logged out automatically. Otherwise, the user can log out at this time without reentering a valid password. Keep in mind the Verify Password

dialog differs from the Bad Logins Allowed field of the User Security Group Configuration window. The Bad Logins Allowed field is also based on the number of invalid log-on attempts allowed, but this field accepts a greater range of values and can be set from 1-100.

UI timeouts only occur if the security group to which the user belongs has a configured UI timeout limit greater than zero. Entering a zero ('0') in the UI Timeout field implies no limit of desktop inactivity.

A session timeout occurs after a duration of desktop inactivity defined in the Session Timeout field. When a session timeout event occurs, the user is automatically logged off the desktop and the user must log on (open the Log In window and provide a valid user name and password combination) to obtain access to the desktop. Any windows open upon a session timeout are closed.

If the user's session expires while the Verify Password dialog is displayed and the user has not provided input, the user is still logged out and the user is no longer able to verify the password. All windows are closed, but the desktop remains open.

Session timeouts only occur if the security group to which the user belongs has a configured session timeout limit greater than zero. Entering a zero ('0') in the Session Timeout field implies no limit of desktop inactivity.

To use UI timeouts in an effective manner, the session timeout must be greater than the UI timeout. This is because if the session timeout is less than the UI timeout, users in the security group will be logged off the desktop (upon a session timeout event) prior to a UI timeout event.

A UI timeout event should occur first so that a user can be alerted and take necessary action to save work. Having a session timeout less than the UI timeout negates the purpose of the UI timeout.

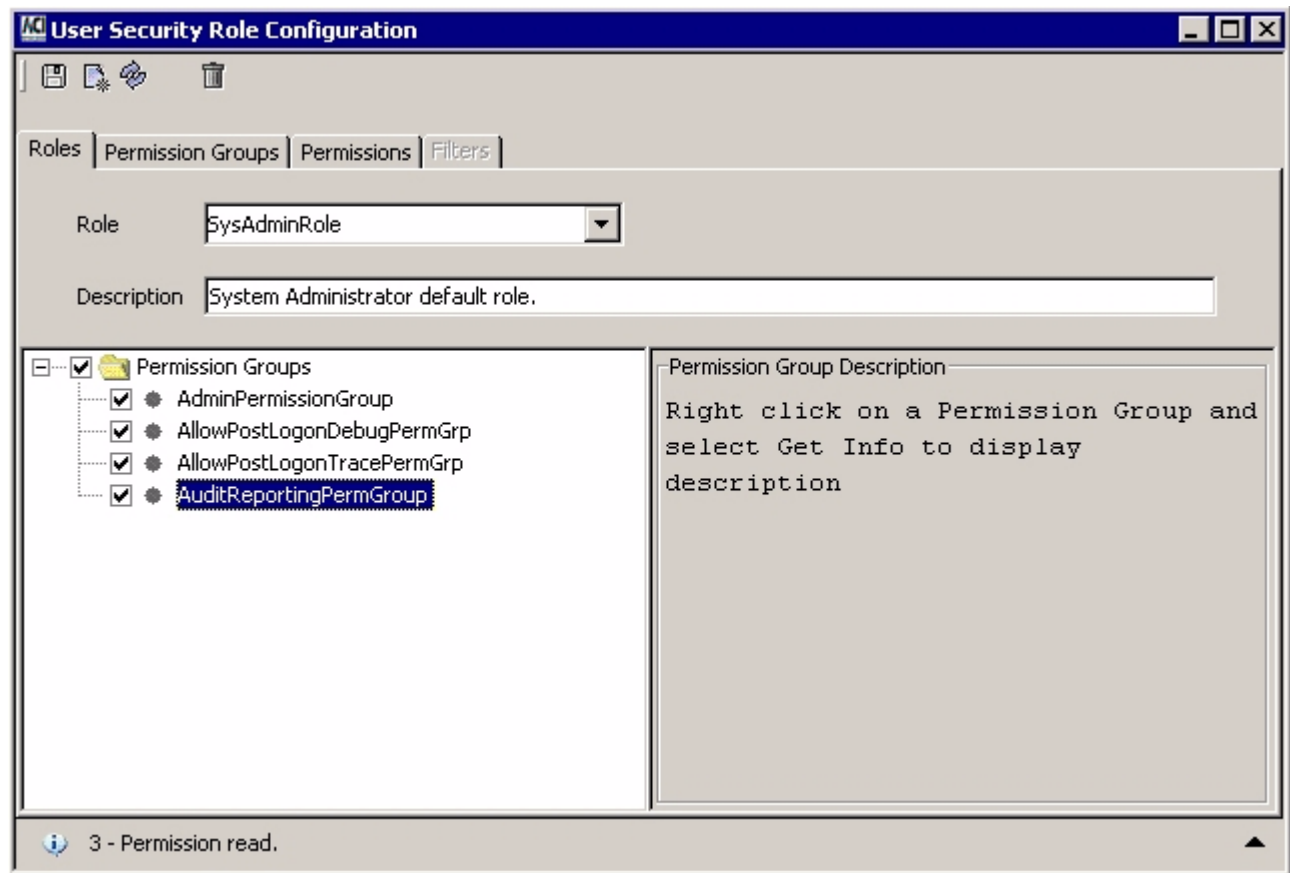
System administration role filter (owner ID)

An existing role ID can be assigned to a security group using the SysAdmin Profile field. Since a role defines a set of permissions (permission groups), tasks IDs and field filters, the role ID at this level limits access to permissions and permission groups that can be configured for users by administrators assigned the security group. To assign a role ID as a filter for a security group, select the role ID in the SysAdmin Profile field.

If a system administration role profile is selected for a security group, only those permission groups, permissions, and tasks defined for the role are accessible to users assigned the security group.

Step 3.0: Configure a user role

To configure a user role, use the Roles tab on the User Security Role Configuration window.



If the available permission groups meet existing needs, enter or select the role ID in the Role field, add or modify the description of the purpose of the role, and select the applicable permissions groups (permission groups can be selected and deselected using the check boxes for each group), then go to Step 4.0: Configure the User's Profile. **Note:** Permissions not defined within your security domain are not available.

If one or more permission groups need to be configured, the following steps might also need to be performed.

- Step 3.1: Configure Field Filters
- Step 3.2: Configure Permissions
- Step 3.3: Configure Permission Groups

Step 3.1: Configure field filters

Note: The functionality of field filtering is the same among applications, but the implementation can vary.

Field filters are configured using the Filters tab on the User Security Role Configuration window, and are applied to a user's profile through permissions.

The screenshot shows the 'User Security Role Configuration' window with the 'Filters' tab selected. The window has a blue title bar and standard Windows window controls. Below the title bar is a toolbar with icons for save, undo, redo, and delete. The main area contains the following elements:

- Four tabs: 'Roles', 'Permission Groups', 'Permissions', and 'Filters' (which is active and highlighted in orange).
- 'Filter ID' and 'Filter Type' dropdown menus.
- A 'Description' text box.
- A 'Filter Value Settings' section containing:
 - A text box for entering a value, followed by an 'Add' button.
 - An 'Available Values' list box on the left.
 - An 'Assigned Values' list box on the right.
 - Four arrow buttons between the list boxes: a single right arrow, a double right arrow, a single left arrow, and a double left arrow.

Note: The Add button and filter value text box is visible only when debugging is enabled and is intended for ACI use only.

Field filters consist of the following elements:

- Unique identification (the Filter ID field)
- The type of filter (the Filter Type field)
- A description of the purpose and use of the filter
- Multiple field values

The values available in the Available Values field are those that have been configured using the applicable configuration window. For example, for the InstitutionID filter type, the values shown in the Filter Value Settings area are those institution IDs that have been configured using the Institution Configuration window.

Tip: Use the asterisk (*) to select all current and future field values.

To add or remove field values, use the following buttons provided on the tab to perform the desired task:

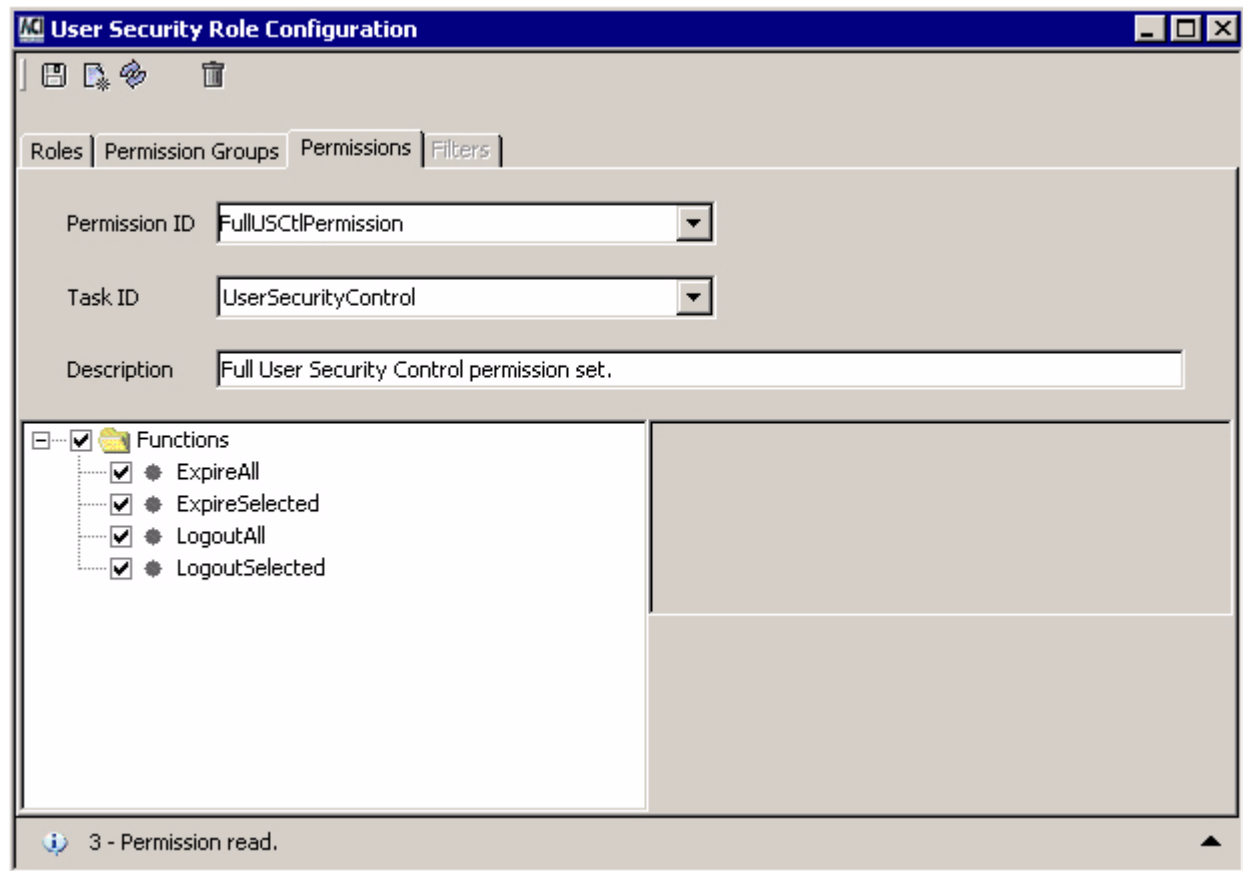
Button	Task
	Move the selected field value(s) from the Available Values field to the Assigned Values field.
	Add all available field values from the Available Values field to the Assigned Values field.
	Remove the selected field(s) value from Assigned Values field back to the Available Values field.
	Remove all selected field(s) values from Assigned Values field back to the Available Values field.

To select a value, click on the value. Selected values have a modified background color. To select multiple values, press down the **Ctrl** key while selecting values.

To add a value not available in the Available Values field, enter the value in the text entry field below the Filter Value Settings area and click the Add button.

Step 3.2: Configure permissions

Use the Permissions tab on the User Security Role Configuration window to create permissions and permission group IDs.



Permission IDs

A new permission ID can be created (entered) or an existing permission ID can be selected. Permissions are available on the Permissions Group tab for selection when configuring a permission group.

The description is to state the purpose of the permission and any special information.

Tasks IDs

Each task ID corresponds to an ACI desktop window.

Note: Task IDs cannot be created using the ACI desktop.

Functions

The functions that a user can perform on a window (task) are defined for each permission on the Permissions tab. A function is selected by entering a check mark in the functions check box. Functions include, but are not limited to, Delete, Save, Send, and View. The available functions vary by task ID.

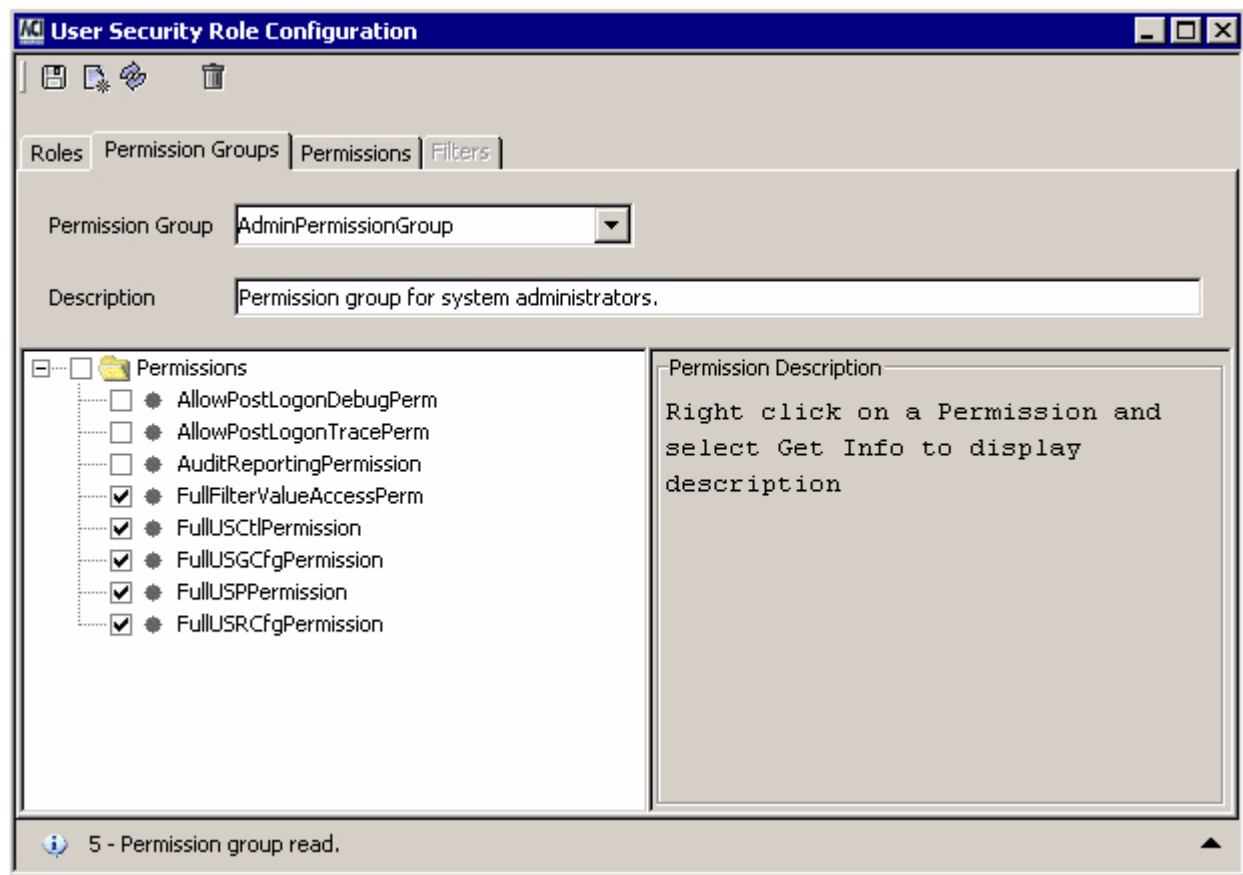
Field filters

If a task ID refers to a window that contains a field or fields that are filterable, the Permissions tab will list the field ID(s) and provide a selectable list of filter IDs that can be selected and applied to the permission.

Note: The actual filterable fields varies by application. See the boxed note in step 3.1 for more information.

Step 3.3: Configure permission groups

To configure a new or existing permission group, use the Permission Groups tab on the User Security Role Configuration window. This tab provides a Permission Group field in which a new permission group ID can be entered or an existing permission group can be selected.



Available permissions, as defined per your security domain, are provided on the tab. Permissions are added or removed from a permission group by selecting (entering a check mark) or deselecting (removing a check mark) in the appropriate checkbox.

Permission information

A description of each permission along with the Task ID assigned to the permission and the allowed functions can be displayed in the Permission Description section of the Permissions Groups tab. To display this information, right click on the desired permission and select Get Info on the pop-up menu.

Step 4.0: Configure the user's profile

To configure a new user (user profile), display the General tab of the User Security Profile Configuration window, enter the user's unique ID in the User ID field and perform the following steps:

- Step 4.1: Set the user's status
- Step 4.2: Assign a security group
- Step 4.3: Enter information about the user
- Step 4.4: Set the user's password
- Step 4.5: Assign roles to the user
- Step 4.6: Assign filters

Note: The User Security Profile Configuration window will not be available when you are using a third-party authenticator other than the one provided with this application. Contact your security administrator for details.

Tip: To add a new user whose profile is based on an existing user profile, display the user profile that is to be used as a template and replace the text in the User ID field with the new user ID. In addition, when the Save button is clicked for a new user, the desktop displays the New User's Password window, which is described in Step 4.4.

Step 4.1: Set the user's status

User status options:

- | | |
|-----------------|-----------------------------|
| Active | = Open and active. |
| Closed/Admin | = Closed without incident. |
| Closed/Inactive | = Closed due to inactivity. |

Disabled/Admin = Disabled, for security reasons, by authorized personnel.
 Disabled/System = Disabled automatically for security reasons (e.g., maximum number of log-on attempts has been exceeded).

Note: This status is set programmatically.

Temporary = User can access the desktop for a predefined amount of time.

Step 4.2: Assign a security group

Security groups are provided in the selectable field next to the Security Group link. If a desired security group is not available, click on the link to access the User Security Group Configuration window to review existing security groups or configure a new security group. Only those security groups defined for your security domain are available.

Step 4.3: Enter information about the user

Information about the user includes:

- Complete name information
- Contact information, such as phones, Fax and email address
- Employment information, such as department, manager and location

Step 4.4: Set the user's password

To set a new user's password, click Save, after which the desktop displays the New User's Password window.

Note: The Password tab of the User Security Profile Configuration window is used for management of existing users. See the section 4 (User management) for more information.

For users assigned a status other than Temporary, the desktop displays the version of the New User's Password window shown below. Indicate whether the user must change the assigned password during the next log-on attempt (or upon the first log-on attempt for a new user), and the start date on which the user can log on to the desktop.

The screenshot shows a Windows-style dialog box titled "New User's Password". It has a standard title bar with a close button. Inside, there are two text input fields: "New Password" and "Verify Password", both containing masked characters (asterisks). Below these fields is a checkbox labeled "Require password change" which is checked. Underneath the checkbox is a section labeled "New User Start" containing a date picker control showing "Thursday, January 11, 2007". At the bottom of the dialog are two buttons: "Ok" and "Cancel".

When a new temporary user (i.e., user status being set to temporary) is created, the desktop displays the New User's Password window, shown below, for indicating whether the user must change the password on logging on, and for setting the start and expiration dates for which the user is active.

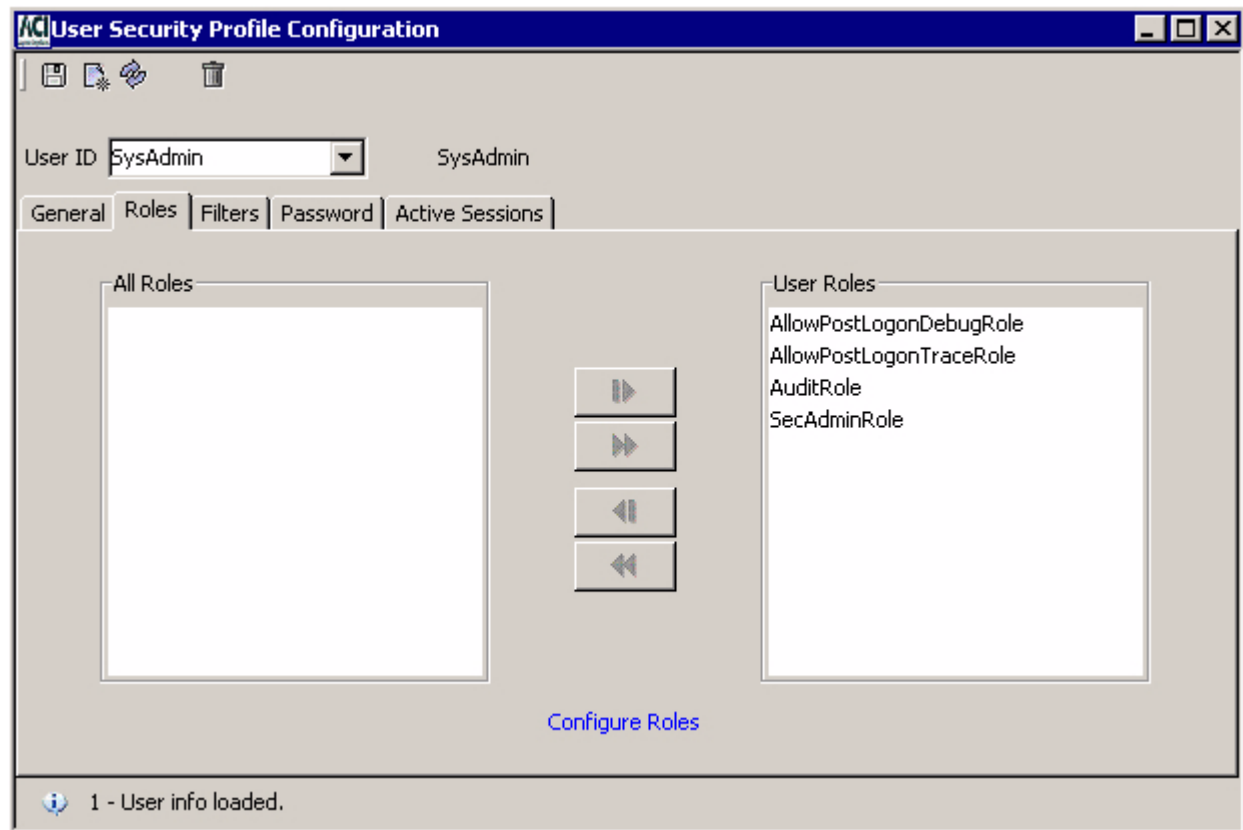


The image shows a Windows-style dialog box titled "New User's Password". It contains two text input fields for "New Password" and "Verify Password". Below these is a checkbox labeled "Require password change". Underneath the checkbox are two date selection fields: "New User Start" (showing "Thursday, January 11, 2007") and "New User Expire" (showing "Friday, January 11, 2008"). At the bottom are "Ok" and "Cancel" buttons.

New User's Password	
New Password	
Verify Password	
☐ Require password change	
New User Start	Thursday, January 11, 2007
New User Expire	Friday, January 11, 2008
Ok	Cancel

Step 4.5: Assign roles

Use the Roles tab on the User Security Profile window to assign (and remove) the roles defined for a user.



This tab displays all the user roles available within your security domain on the ACI desktop. Roles not defined within your security domain are not displayed. User roles assigned to a user are shown in the User Roles field. Available user roles not assigned to a user are shown in the All Roles field. To add or remove user roles from a user profile, select the role or roles and use the buttons provided on the tab to perform the desired task. The buttons are:

Button	Task
	Add the selected role(s) to the user profile (move from the All Roles field to the User Roles field).
	Add all available roles to the user profile (move from the All Roles field to the User Roles field).
	Remove the selected role(s) from the user profile (move from the User Roles field to the All Roles field).

Button	Task
	Remove all assigned roles from the user profile (move from the User Roles field to the All Roles field).

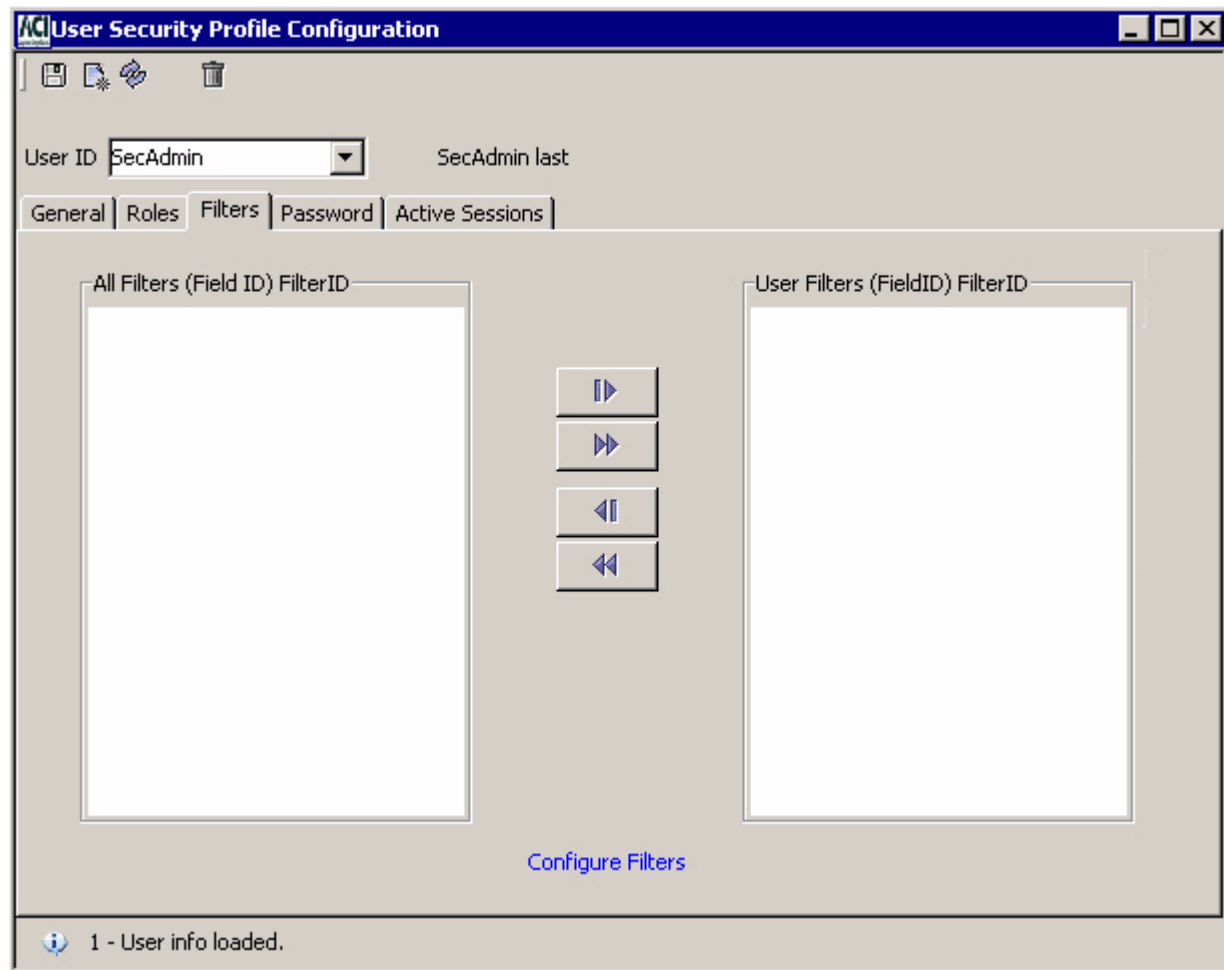
To select a role, click on the role. Selected roles have a modified background color. To select multiple roles, press down the **Ctrl** key while selecting roles.

If a particular role is not available or an existing role needs to be changed, click on the Configure Roles link to access the User Security Role Configuration window to configure a new or existing role.

Note: A user cannot change the role configuration for the profile (User ID) that was used to log on. In such situations, when the Roles tab is accessed for that User ID, the All Roles field and User Roles fields are inactive.

Step 4.6: Assign filters

Use the Filters tab on the User Security Profile window to assign (and remove) the filters defined for a user.



To add or remove filters from a user profile, select the filter or filters and use the buttons provided on the tab to perform the desired task. The buttons are:

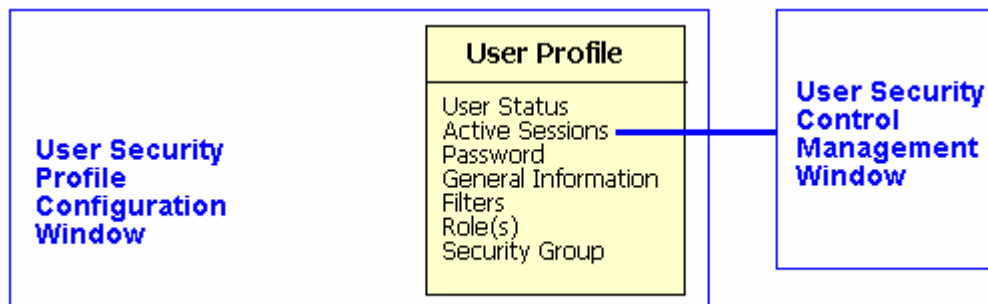
Button	Task
	Add the selected filter(s) to the user profile (move from the All Filters (Field ID) Filter ID field to the User Filters (FieldID) Filter ID field).
	Add all available filters to the user profile (move from the All Filters (Field ID) Filter ID field to the User Filters (FieldID) Filter ID field).

Button	Task
	Remove the selected filter(s) from the user profile (move from the User Filters (FieldID) Filter ID field to the All Filters (Field ID) Filter ID field).
	Remove all assigned filters from the user profile (move from the User Filters (FieldID) Filter ID field to the All Filters (Field ID) Filter ID field).

4: User management

Security administrators can set the status, display and terminate active sessions, and expire passwords by password group of users within the administrator's security domain. This activity is performed from the user security windows as described in this section.

The illustration below shows the user profile and includes the name of the applicable windows used to manage user profiles.



The User Security Profile Configuration window is used for configuring a user profile; however, it can be used to view a user's active sessions, to change an existing user's status and to manage a user's password.

The User Security Control Management window is used to view and manage the active sessions for all users. The window can also be used to expire passwords by password group.

To assist system administrators and other authorized users, two additional windows are available and are discussed in this section:

- User Status window
- User Permissions window

Set the status of a user

A user's status can be changed automatically if security and password group requirements have been violated. A user's status can also be changed by authorized users.

Use the General tab on the User Security Profile Configuration window to view and set a user's status.

The screenshot shows the 'User Security Profile Configuration' window with the 'General' tab selected. The 'User ID' is 'systemUser01'. The 'User Status' dropdown is open, showing options: Active, Closed/Admin, Closed/Inactive, Disabled/Admin, Disabled/System, and Temporary. The 'Security Group' is 'Level1 SecurityGroup'. Other fields include First Name, Middle Name, Last Name, Department, Location, Manager, Email, Work Phone, Mobile Phone, Fax, Create Time (1/11/07 1:00 PM), Last Login (6/12/07 12:43 PM), Last Failed Login (4/18/07 7:58 AM), and Bad Login Tries (1).

Authorized users can assign the following status codes to a user:

Active	= Open and active.
Closed/Admin	= Closed without incident.
Closed/Inactive	= Closed due to inactivity.
Disabled/Admin	= Disabled, for security reasons, by authorized personnel.
Disabled/ System	= Disabled automatically for security reasons (e.g., maximum number of log-on attempts has been exceeded).
Temporary	Note: This status is set programmatically. = User can access the desktop for a predefined amount of time.

Active user session management

When a user has logged on to the desktop by providing a valid user ID and password, an active session is created for the user. An active session contains the following information:

- The user's ID.

- The unique session number assigned to the user's session.
- The workstation ID, such as a network ID or URL, from which the user has logged on to the desktop.
- The date and time the user began the session.
- The date and time the current session expires.

Information about active user sessions is available on two separate Active Sessions tabs. One tab is provided on the User Security Profile Configuration window and displays the active sessions for a selected user. The other Active Sessions tab is located on the User Security Control Management window and displays active user sessions for all users within a security administrator's security domain (security group to which the administrator has access).

Active User Session for a Selected User:

The screenshot shows the 'User Security Profile Configuration' window. At the top, there is a 'User ID' dropdown menu set to 'systemUser01'. Below this are five tabs: 'General', 'Roles', 'Filters', 'Password', and 'Active Sessions'. The 'Active Sessions' tab is selected. It contains a table with two columns: 'Workstation ID' and 'Logged in time'. The table has one row with the values 'Network/URL' and '2007-6-12 12:43:11.39'. Below the table, there is a 'Password' label, a blue link 'All Active Sessions', and a 'Logout Selected' button.

Workstation ID	Logged in time
Network/URL	2007-6-12 12:43:11.39

Active User Sessions for All Users:



Note: The active sessions of the application can change at any time; therefore, to update the list of active sessions, click Refresh ().

Terminate active user sessions (log off a user)

One or more active user sessions can be terminated (i.e., logged off) from either one of the two windows that are used to display active user sessions — the User Security Profile Configuration window and the User Security Control Management window. To log off one or more user sessions, display and select the user session(s) and click Logout Selected.

Password management

Authorized users can view a user's password change history, add or change a user's password, and expire the user's current password by accessing the Password tab on the User Security Profile window and selecting the user's ID.

To change a user's password, display the Password tab on the User Security Profile Configuration window and select the user ID. Enter the new password in both the New Password and Verify Password text boxes and click Save (). If the change is successful, a confirmation message appears in the status pane of the window. Otherwise, an appropriate error message is displayed.

To expire a user's password, display the Password tab on the User Security Profile Configuration window, select the user ID, and click Expire Password. If a user's password has been expired but a new password has not been defined, the user is prompted to create a new password on the next log-on attempt.

Authorized users can access a table that shows the password groups that have been assigned to security groups, and can expire all user passwords of users assigned to a password group.

Use the Password Maintenance tab on the User Security Control Management window to view password group usage and expire password groups.

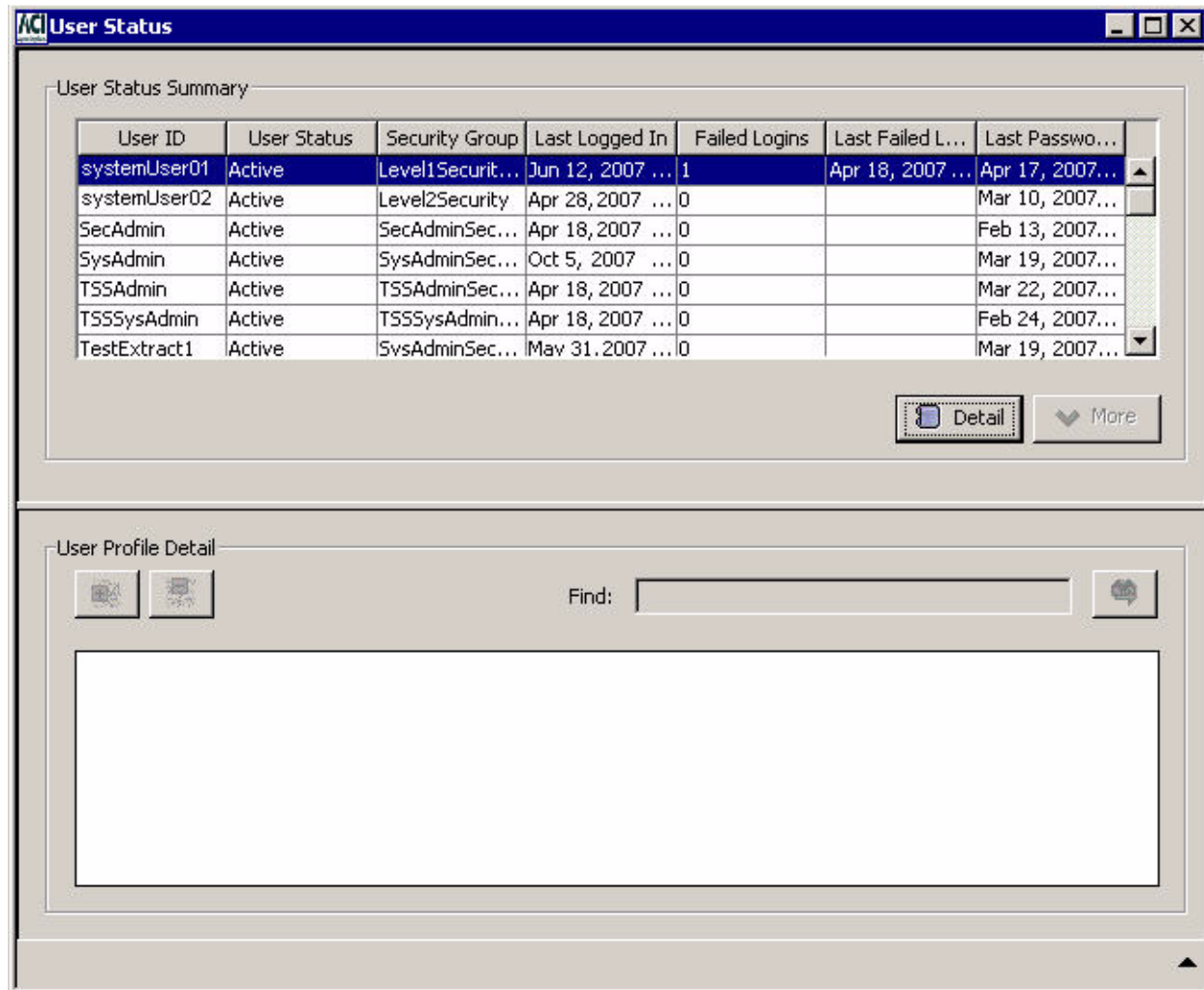
A summary of password changes for all users can be displayed on the User Status window (**View > User Status**) as discussed on the following page.

User Status window

The User Status window (**View > User Status**) lists summary information for users.

Note: The User Status window will not be available when you are using a third-party authenticator other than the one provided with this application. Contact your security administrator for details.

Example User Status window:



When this window is accessed, a list of user IDs with user role and login information is displayed.

This window allows quick identification of users with specific status exceptions and stale (i.e., idle) accounts. To identify such users, if the **More** button is active, click the button to retrieve additional users. The button remains active until all user accounts have been retrieved. The users can be sorted by login information by clicking the **Last Logged In** column. Other columns can be used to sort the users by different criteria. A secondary sort can be obtained by holding down the CTRL key and selecting another column header.

To display a detail report for a user, select the row of the user and click **Detail**. The summary information displayed in the text area can be printed using the Print function available on the File menu.

User Status Detail properties

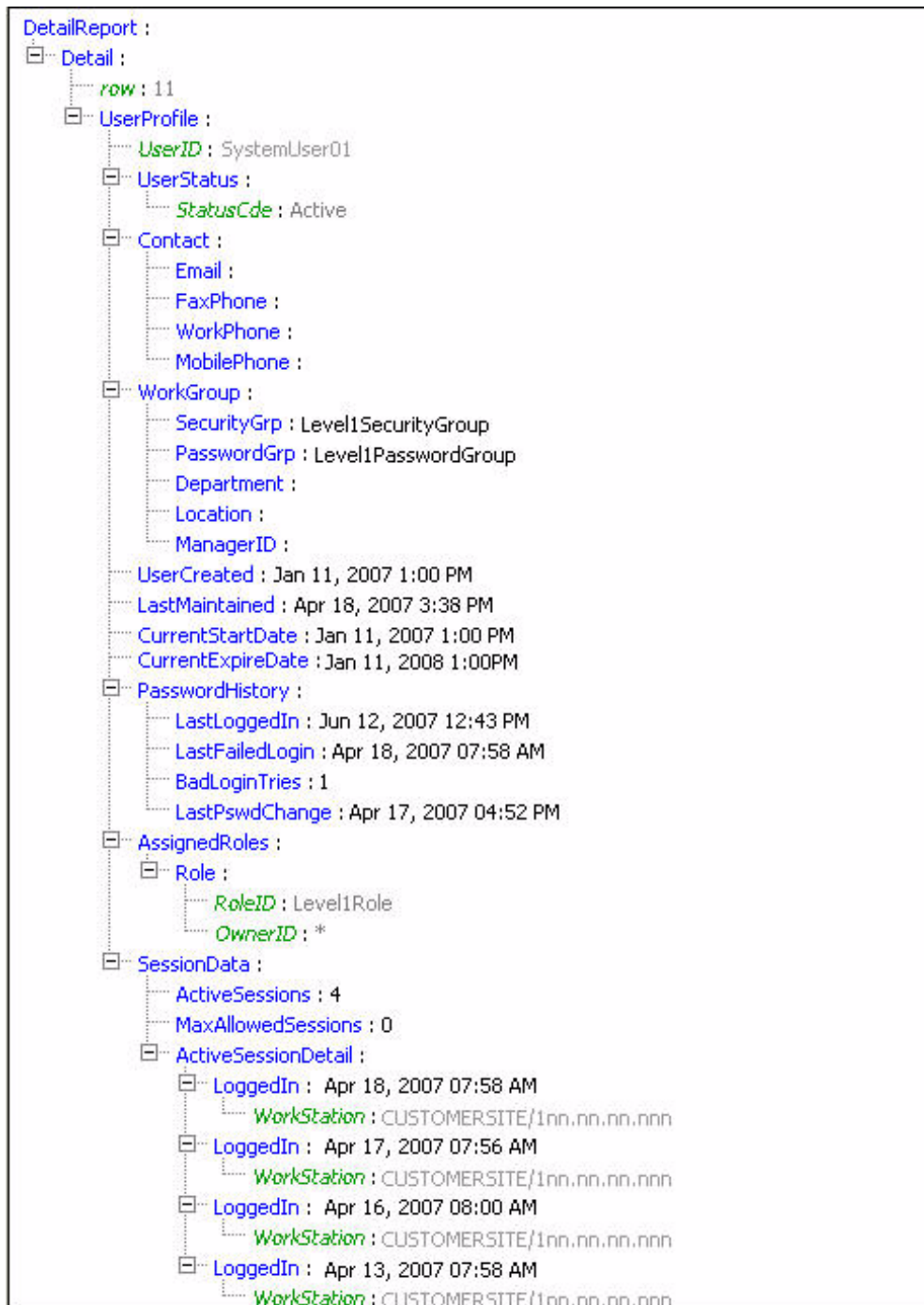


In support of PCI compliance, a recurring report of the information shown on this window can be configured using the following properties documented in the ***BASE24-eps Java server reference guide***:

report.userdetail.loc	report.userdetail.name
report.userdetail.timeperiod.hours	report.userdetail.starttime

User Status Detail example

Below is an example of the detail information available on the User Status window.

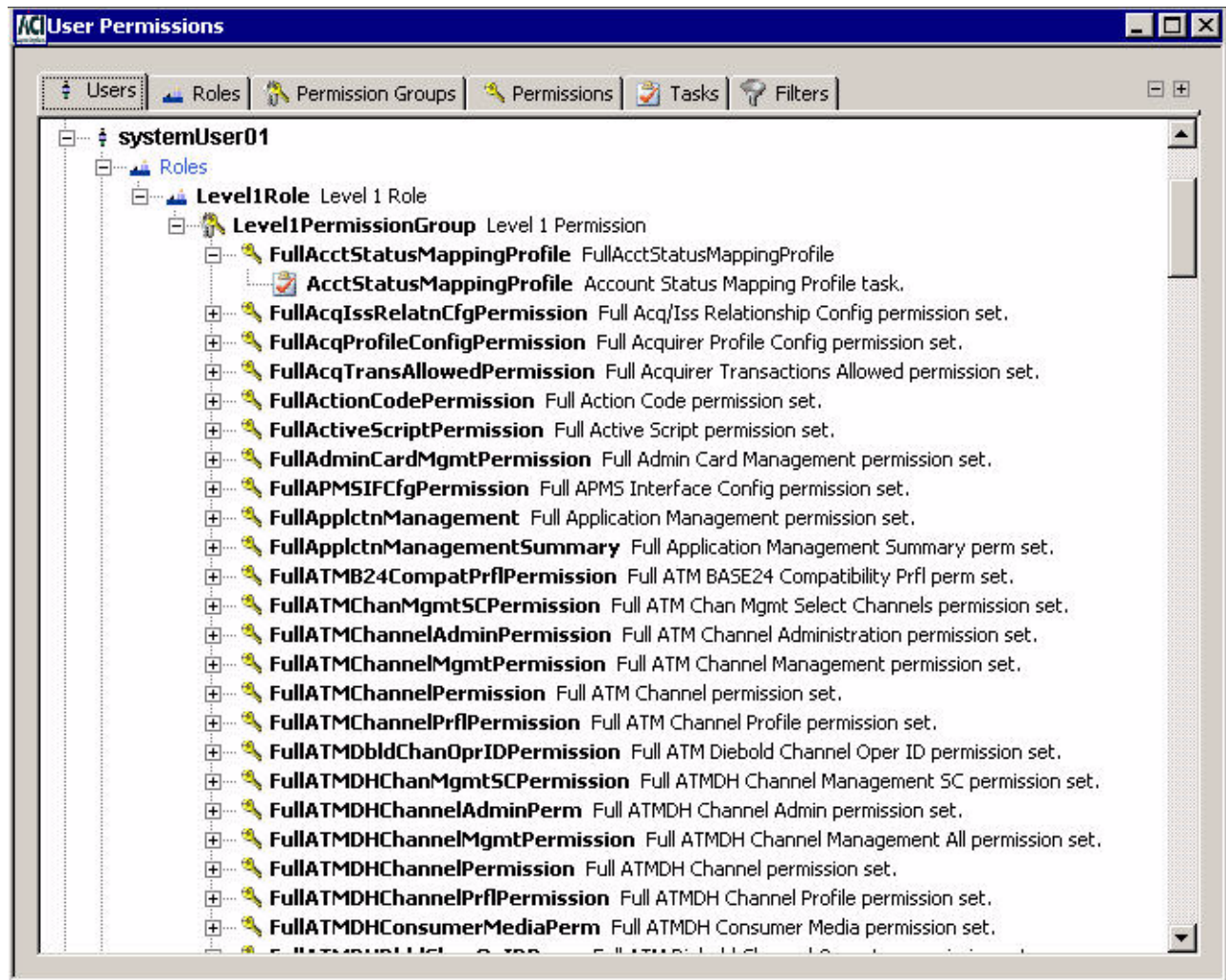


User Permissions window

The User Permissions window (**View > User Permissions**) provides hierarchical information for users, roles, permission groups, permissions, tasks, and field filters.

Note: The User Permissions window will not be available when you are using a third-party authenticator other than the one provided with this application. Contact your security administrator for details.

Example User Permission window:



When this window is accessed, hierarchical user security information lists are provided and accessible on the various tabs of the window. For example, select the Tasks tab to identify all task IDs and the role, permissions, permission groups, and users assigned a given task.

Note: Distributed Security Administration filters do not apply to the information displayed on the User Permissions window. Therefore, user security access to the window should only be provided to System-level security administrators.

User Status Detail properties



In support of PCI compliance, a recurring report of the information shown on this window can be configured using the following properties documented in the ***BASE24-eps Java server reference guide***:

report.userpermissions.loc	report.userpermissions.name
report.userpermissions.timeperiod.hours	report.userpermissions.starttime

5: User auditing

User audit is the logging of user activity to a data source for the purpose of reporting and monitoring the user activity.

Whenever a user of the desktop initiates and completes an activity (e.g., log in, log out, change password, delete, or save) from a window, a message that contains information about the user activity is written to the User Audit Log data source.

The audit message sent to the log data source can be one of the following:

- Header information only about the changes to affected records
- Header information and detail information about the before and after data images of affected records

The application server writes audit messages directly to the USRAULOG data source or to the Audit_Store_and_Forward data source depending on the middle ware capabilities of the hardware platform.

For information about enabling user auditing and setting the audit level for a desktop window, refer to the **BASE24-eps Java server reference guide** for information about audit properties, such as:

- db.audited
- db.audited.*table-name*, where *table-name* is the physical table name

Audit report XML format

The information stored in the extracted audit reports is in XML format. The following example is an XML-formatted message containing user activity information.

```
<?xml version="1.0" encoding="UTF-8" ?>
<AuditReport>
  <AuditDetails row="0">
    <Audit_LoggingRq>
      <Timestamp>TimeStamp Info</Timestamp>
      <ContextDescr>InsertRq</ContextDescr>
      <Ticket>
        <UserID>User Profile ID</UserID>
```

```

        <SessionNumber>1</SessionNumber>
        <TerminalID>OMAUSER/ip_addr</TerminalID>
    </Ticket>
    <ActivityGroup>
        <ActivityType>Table Function</ActivityType>
        <TableGroup>
            <TableName>Table</TableName>
            <KeyGroup>
                <Column>
                    <ColumnName>Table Element</ColumnName>
                    <BeforeValue />
                    <AfterValue>New Value</AfterValue>
                </Column>
                <Key>
                    <KeyColumn>
                        <ColumnName>Key</ColumnName>
                        <AfterValue>New Value</AfterValue>
                    </KeyColumn>
                </Key>
            </KeyGroup>
        </TableGroup>
    </ActivityGroup>
</Audit_LoggingRq>
</AuditDetails>
</AuditReport>

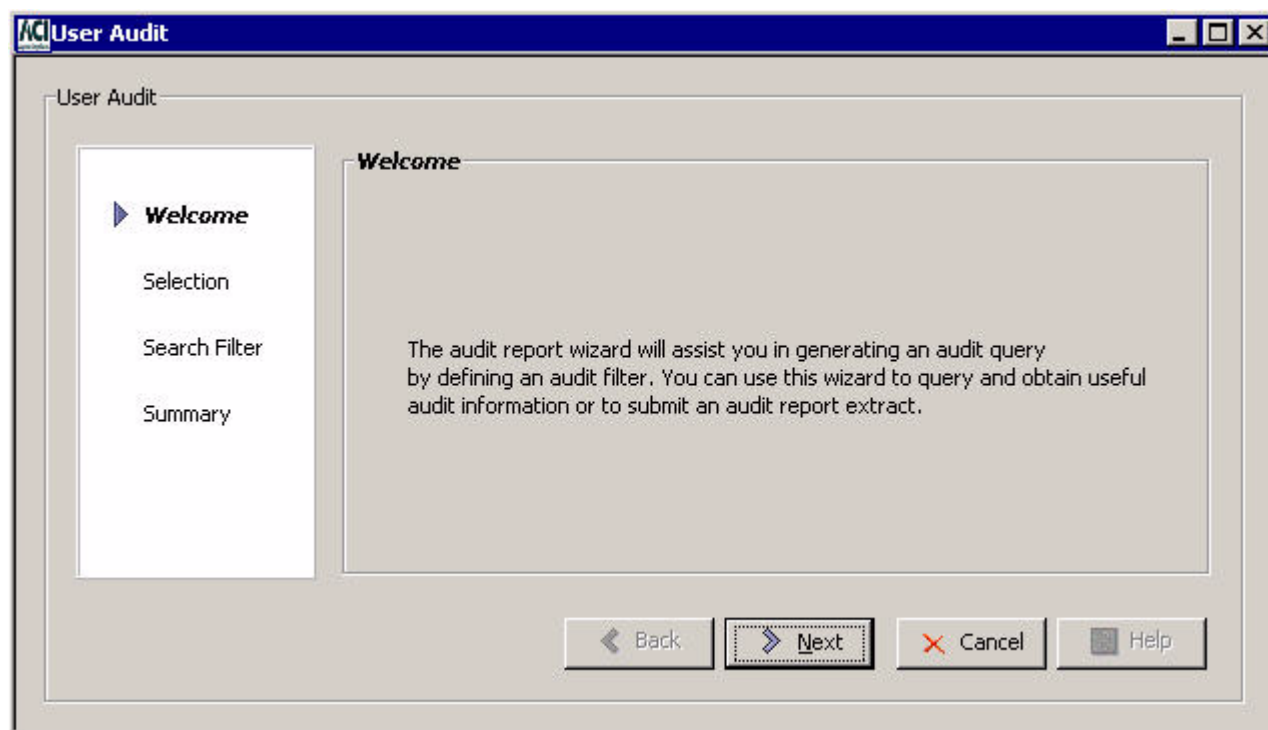
```

The root element of the XML message is <AuditReport> that includes one or more <AuditDetails></AuditDetails> sections. <AuditDetails> contains important information: a time stamp (<Timestamp>), Context information (<ContextDescr>), ticket information (<Ticket>), and activity information (<ActivityGroup>). Information related to the user and the user's session is included in the ticket. Information about the activity performed and the data source (table) affected is included in the activity group.

Note: Use an XML reporting tool such as XML Spy, MS-Access 2002, or Crystal Reports to evaluate and format the XML information as well as to generate reports.

User audit activity report process

Activity reports for selected users can be initiated by accessing User Audit on the View menu. User activity is extracted from the USRAULOG data source for a specified user ID between a start date and time and an end date and time. This information can be viewed on the ACI desktop or sent to a designated file.



The reports provide the following information:

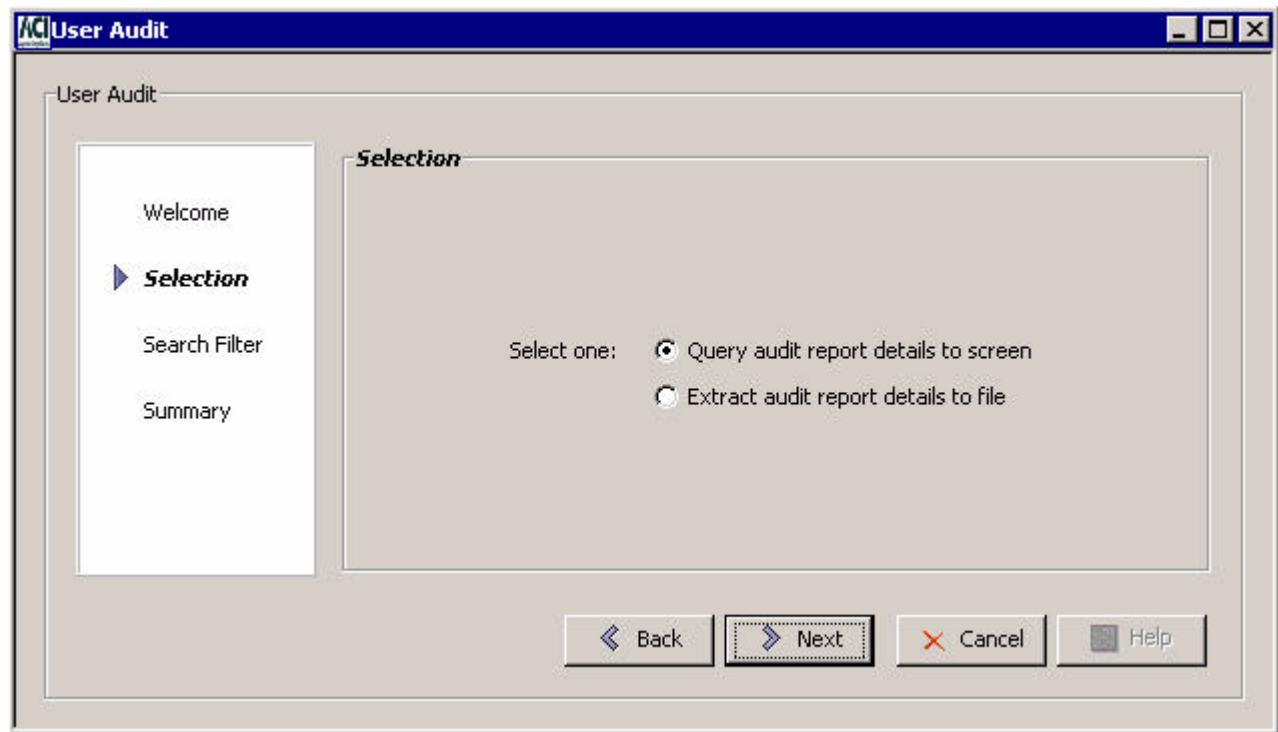
- Date and time of action.
- User ID performing the action.
- Terminal location from which action was performed.
- Type of action performed.
- File and field name affected by the action, if applicable.
- Before and after image of data modification action, if full auditing is set.

The location where the report file is saved is specified by the **report.userauditlog.loc** environment attribute, which defines the directory location. See the previous page for an example report file.

Click **Next** to proceed.

Selection

Below is the Selection step of the User Audit function.



1. Select whether to have the report sent to the screen or written to a file (the file name is specified in the next step window).
2. Click **Next**.

Search filter

Below is the Search Filter step of the User Audit function.

The screenshot shows a window titled "User Audit" with a sidebar on the left containing four options: "Welcome", "Selection", "Search Filter" (which is highlighted with a blue arrow), and "Summary". The main area of the window is titled "Search Filter" and contains the following fields and instructions:

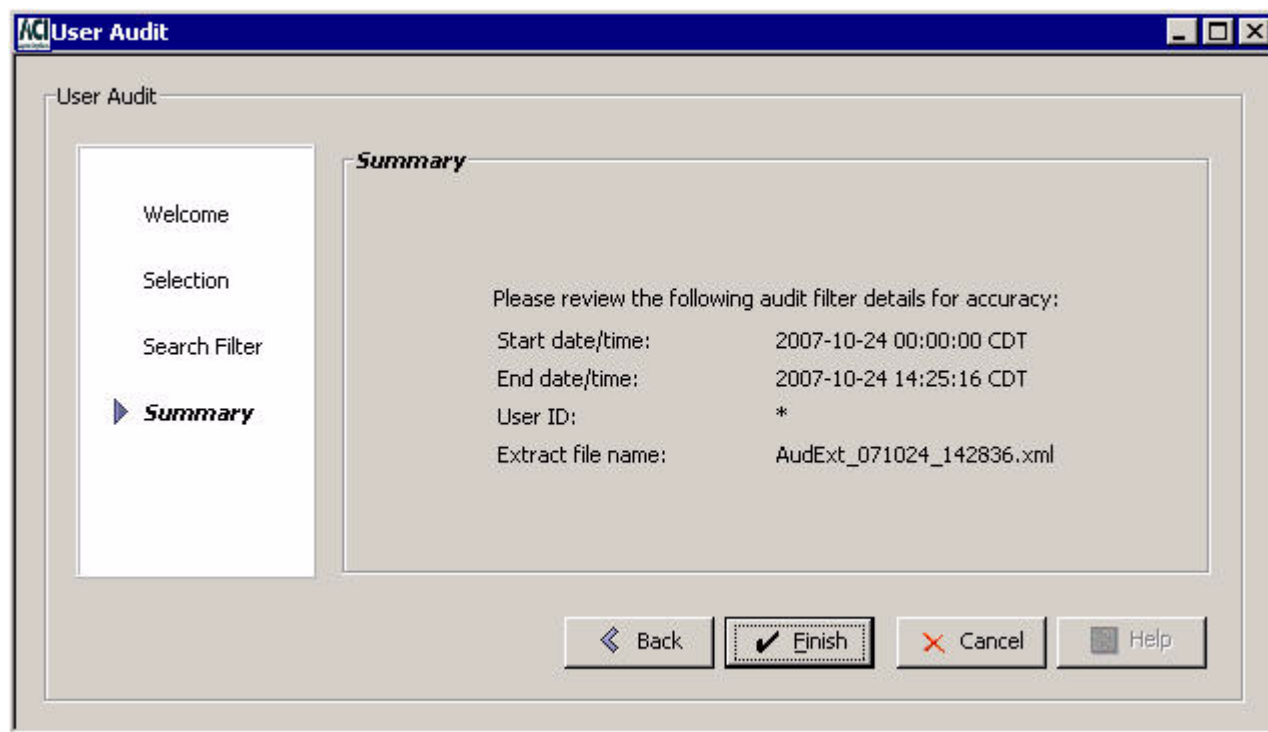
- Start date/time: A date field set to "October 24, 2007" and a time field set to "12:00:00 am CDT".
- End date/time: A date field set to "October 24, 2007" and a time field set to "02:25:16 pm CDT".
- User ID: A text field containing an asterisk (*). Below it, a note says "(Specify '*' to select all users.)".
- Extract file name: A text field containing "AudExt_071024_142836.xml". Below it, a note says "(The extracted report will be stored on the server.)".

At the bottom of the window, there are four buttons: "Back", "Next" (which is highlighted with a dashed border), "Cancel", and "Help".

1. Select the start and end date/time fields.
2. Enter a specific user ID or an asterisk (*) to report on all users.
3. Select the file name of the extract XML report file.
4. Click **Next**.

Summary

Below is the Summary of the User Audit function.



1. Review the summary information provided.
2. If the summary information is incorrect, click **Back** to step back through the process to make changes.
3. If the summary information is correct, click **Finish** to begin building the report, whether for screen or written to file.

Example report on the ACI desktop

When a report is completed, the results are shown in the Audit Report Viewer for which an example is shown on the following page.

If the **More** button is active, click the button to retrieve additional users. Note that the **More** button remains active until all user accounts have been retrieved. The user information displayed in the table can be sorted using the column headers. Additional column headers can be selected by pressing the **Ctrl** key.

To display detail information for a user, select a summary role and click the **Detail** button. Double clicking in a summary row also initiates the display of detail information for a user.

Audit Report Viewer

Audit Summary

#	Local Date/Time	User ID	Workstation	Activity Type	Activity Detail	Result	Service
1	Sep 21, 2007 1...	systemuser01	oma2ui1	Logoff	USEC		
2	Sep 21, 2007 2...	systemuser	SystemProcess	View	SecurityReport...		USERSECURIT...
3	Sep 21, 2007 ...	systemuser03	oma2ui1	Logon	LoginComma...	OK	USERSECURIT...
4	Sep 21, 2007 2...	SysAdmin	oma2ui1	View	ISOMsgPrflistRq		
5	Sep 21, 2007 2...	SecAdmin	oma2ui1	View	InterfaceListR...		
6	Sep 21, 2007 2...	systemuser07	oma2ui1	View	InstConfigGetI...		

Detail More

Audit Detail

Find:

AuditReport :

- AuditDetails :
 - row : 3
 - Audit_loggingRq :
 - Timestamp : 2007-09-21T07:30:11.734Z
 - ContextDescr : LoginCommand
 - Ticket :
 - UserID : systemuser03
 - SessionNumber : 48
 - TerminalID : oma2ui1/172.17.1.177
 - ErrorData : OK
 - ActivityGroup :
 - ActivityType : Logon

Automated user auditing

User audit extracts to file can be performed automatically by specifying the `com.aciworldwide.common.timertask.UserAuditReport` classname for the configuration properties file `timertask.function.n.classname` configuration parameter. The Java Server Framework application specified as the `timertask.manager.processid` invokes the classname at application initialization.

The following properties are used to configure automated user audit reporting:

Property	Description
<code>report.userauditlog.loc</code>	The file path for the user audit report.
<code>report.userauditlog.name</code>	The name of the user audit report.
<code>report.userauditlog.offset.days</code>	The number of days prior to the current day to use for obtaining user activity for the report.
<code>report.userauditlog.report.starttime</code>	The time at which the report is generated.

For example, if today is Wednesday and the time is 9:00 AM, the offset is one day, and the report start time is 2:00 AM, then the most recent audit information available is that for 2:00 AM Tuesday to 2:00 AM Wednesday.

Refer to the ***BASE24-eps Java server reference guide*** for more information about the above properties.

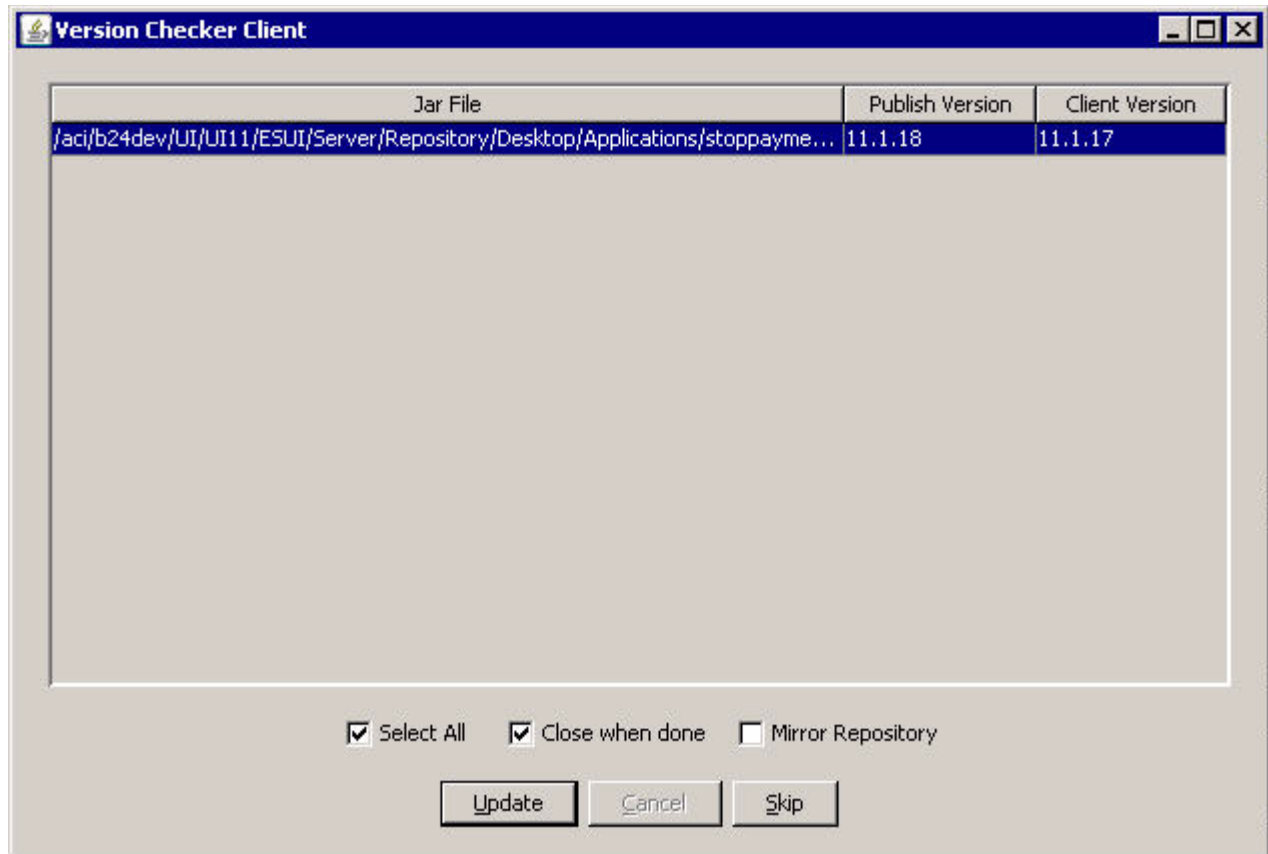
6: Version Checker

The desktop provides a way for clients to keep the desktop configuration current. This configuration support is provided by the Version Checker, which is a utility that manages the versions of Java™ Archive (JAR) files of desktop components that are distributed to ACI desktop clients. The Version Checker automatically runs during startup of the desktop at which time it compares the current version of the client files to new versions, which reside in a predetermined location (i.e., repository). The location of the Version Checker repository is defined during installation. New files that are to be distributed to clients are placed in this location.

The Version Checker automatically runs when a desktop is started per commands located in the **RunDesktop.bat** file. Desktop shortcuts specified during installation point to this batch file. Whether a user is prompted to accept any new files or the files are automatically downloaded is dependent upon configuration settings as discussed in this section.

The files on the repository for which the version differs from that of the client are displayed in the Version Checker Client window. This window provides a list of the files along with the version number of each file on the repository and the version number of the file on the client. If automatic file updating is enabled (discussed later in this section), then this window is not displayed.

To download new files from the repository, select the applicable files and click Update. This update process can be skipped or canceled.



The **Select All** option can be used to select all listed files on the Version Checker Client window. Individual files can be selected by deselecting the Select All option and then clicking in the row for each desired file to download. Only selected files are downloaded.

If the **Close When Done** option is selected, the Version Checker Client window will automatically close after completion of the download of all selected files.

By selecting **Mirror Repository** on the Version Checker Client window, verification is initiated to ensure that no jars reside on the client that are not also on the server repository. That is, any files that exist on the client that do not exist on the current version of the server repository are deleted from the client.

The Version Checker can be configured to download updated and new files automatically.

Including the parameter “-Q” in the Version Checker command line located in the **RunDesktop.bat** file enables automatic file updating.

For example:

```
call "C:\ACI\VersionChecker\verChk" -Q
```

7: Online help

The ACI desktop provides online help for windows (screens) and window components (e.g., fields). This help is displayed in a Java-based HTML browser, called the **Helpset Viewer**:



The components of the Helpset Viewer are:

- 1 - The **Title Bar** in which the name of the helpset is displayed.
- 2 - The **Toolbar** that contains various menu and button functions.
- 3 - The **Navigation Pane** in which are located Contents and Index tabs useful for navigating to desired help content, and a Search tab for searching content.
- 4 - The **Content Pane** in which help content is displayed.
- 5 - The **Contents** tab that provides structured access to window-level help files.
- 6 - The **Index** tab that provides alphabetical access to all help topics.
- 7 - The **Search** tab for searching help content.

Access to JavaHelp from the ACI desktop

JavaHelp content can be accessed with or without context. These two methods are explained below.

With context

Accessing online help with context displays the Helpset Viewer with content displayed in the Content Pane that is specific to a window or window component. To display help with context from the ACI desktop, select **Help > Context Help** or click  on the toolbar. Either of these actions display the Help Pointer  to use for clicking in a general area of a window for window-level help or on a window field for field-level help.

If a window is not open, help with context is not available because an active window in the Application Area of the ACI desktop is necessary to provide context. When context help is not available, the Context Help function and the  button are inactive.

Note that the traditional way to access field-level help by placing the cursor in the field and pressing the **F1** key can also be used.

Without context

To display online help without context from the ACI desktop, select **Help > Help Topics** from the Menu Bar or click  on the Toolbar. A window does not need to be open to access online help without context. When online help is accessed using this method, the Helpset Viewer is displayed with the default home page provided in the content pane and the default TOC view in the navigation pane.

If the Help Topics function and the  button are not enabled, then the ACI desktop does not have access to online help. Check with a System Administrator to obtain ACI online help.

JavaHelp files

The following files together are referred to as a **JavaHelp helpset**:

- The **helpset file** (ACI_Help.hs), which is identified by the .hs file extension, defines the name of the helpset displayed in the Title Bar, the JavaHelp components used, the name of the map file, the names of the table of contents and index files, and the ID of the default content file to display in the Content Pane when help is accessed without context. The helpset file is the top-level file in the JavaHelp helpset.

Be careful not to confuse the term “JavaHelp helpset” with the helpset (.hs) file. *JavaHelp helpset* refers to a collection of JavaHelp files as discussed on this page. Whereas the helpset (.hs) file is a single file (see following page).

- The **map file** is an XML file identified by the .jhm file extension that corresponds identifiers to HTML content files. This file is necessary to provide context-sensitive help for windows and window components as discussed later in this section.
- The **table of contents file** that defines the structure and links of the text entries of the Contents tab.
- The **index file** that defines the structure and links of the text entries of the Index tab located in the Navigation Pane of the Helpset Viewer.
- **HTML files** in which help content resides.
- **Cascading Style Sheets (CSS)** define the style rules applied to content.
- The **images** displayed within HTML content files. Images used in content are in Graphical Interchange Format (GIF) and Joint Photographic Experts Group (JPEG or JPG) formats.

JavaHelp packages

ACI online help is packaged and distributed in the following Java™ Archive (JARs) files:

- **ACI_Help_apf.jar** — For APF-Admin ACI desktop windows
- **ACI_Help_es.jar** — For BASE24-eps
- **ACI_Help_tss.jar** — For Transaction Security Services (TSS)
- **ACI_Help_pm.jar** — For ACI Payments Manager (PM)

Helpset file

The helpset file defines the Java components used, home page, map file, TOC, index, and views. Below is an example helpset file. Comments between “<!--” and “-->” have been added to provide additional information.

```
<?xml version='1.0' encoding='ISO-8859-1' ?>
<!DOCTYPE helpset
  PUBLIC "-//Sun Microsystems Inc.//DTD JavaHelp HelpSet Version 2.0//EN"
  "http://java.sun.com/products/javahelp/helpset_2_0.dtd">
<helpset version="2.0">

  <!-- The title displayed in the Title Bar -->
  <title>ACI Online Help</title>

  <!-- Map information -->
  <maps>
    <homeID>splash</homeID> <!-- The map ID of the splash page -->
    <mapref location="map.jhm" /> <!-- The map file -->
  </maps>

  <!-- Each view represents a tab on the Helpset viewer -->
  <!-- This helpset has 3 views. -->
  <!-- Immediately below is the TOC view -->
  <view>
    <name>TOC</name>
    <label>Contents</label> <!-- The popup (flyover) text of the tab -->
    <type>javax.help.TOCView</type> <!-- The JavaHelp TOC component -->
```



```

        <data>toc.xml</data> <!-- The name of the table of contents file -->
    </view>

    <!-- This is the index view -->
    <view>
        <name>Index</name>
        <label>Index</label>
        <type>javax.help.IndexView</type> <!--
        <data>ix.xml</data> <!-- The name of the index file -->
    </view>

    <!-- This is the search view -->
    <view>
        <name>Search</name>
        <label>Search</label>
        <type>javax.help.SearchView</type>
        <data engine="com.sun.java.help.search.DefaultSearchEngine">
            JavaHelpSearch
        </data> <!-- The search engine defined by the engine attribute -->
        <!-- And, the search database. In this case, the search -->
        <!-- database is located in a folder named JavaHelpSearch -->
    </view>
</helpset>

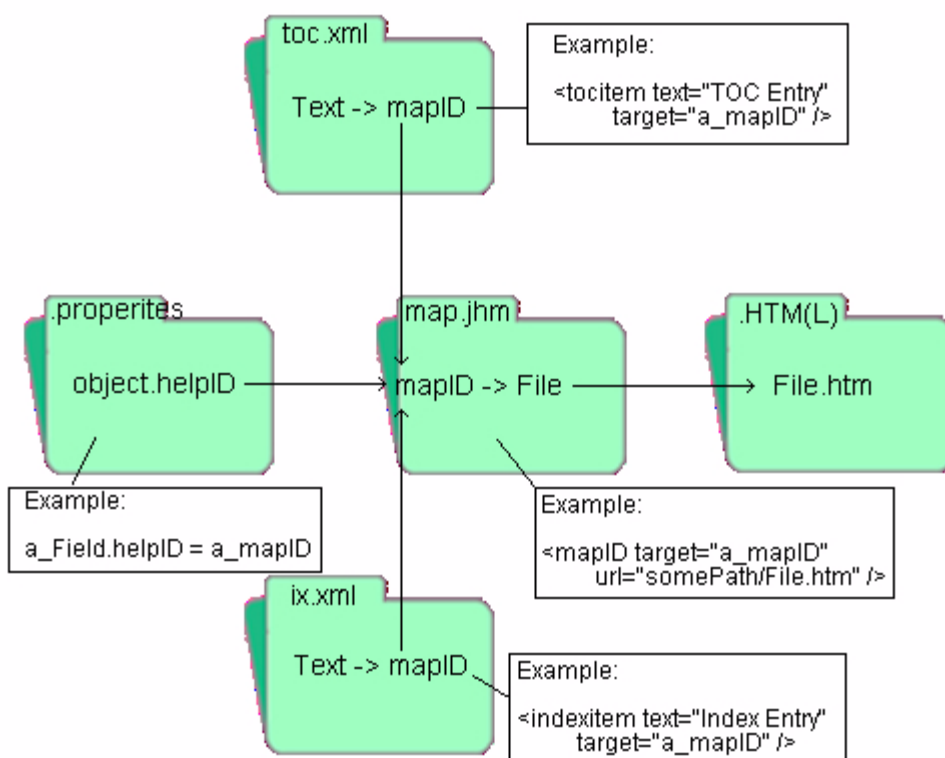
```

How context is implemented in JavaHelp

Help context is implemented in JavaHelp by the use of unique identifiers. Each identifier (referred to as a map ID or help ID) appears as necessary in the following files:

- The property (.properties) file for a window (desktop application)
Note: Most application windows have their own properties file. Also, some properties common among all windows are located in common property files that are shared among the window properties files.
- The JavaHelp map file, map.jhm (pmdesktop.jhm for ACI Payments Manager)
- The table of contents file, toc.xml (ACI_Help_pmtoc.xml for ACI Payments Manager)
- The index file, ix.xml (all applications)

The use and relationship of the identifiers in the above files is shown in the following illustration.

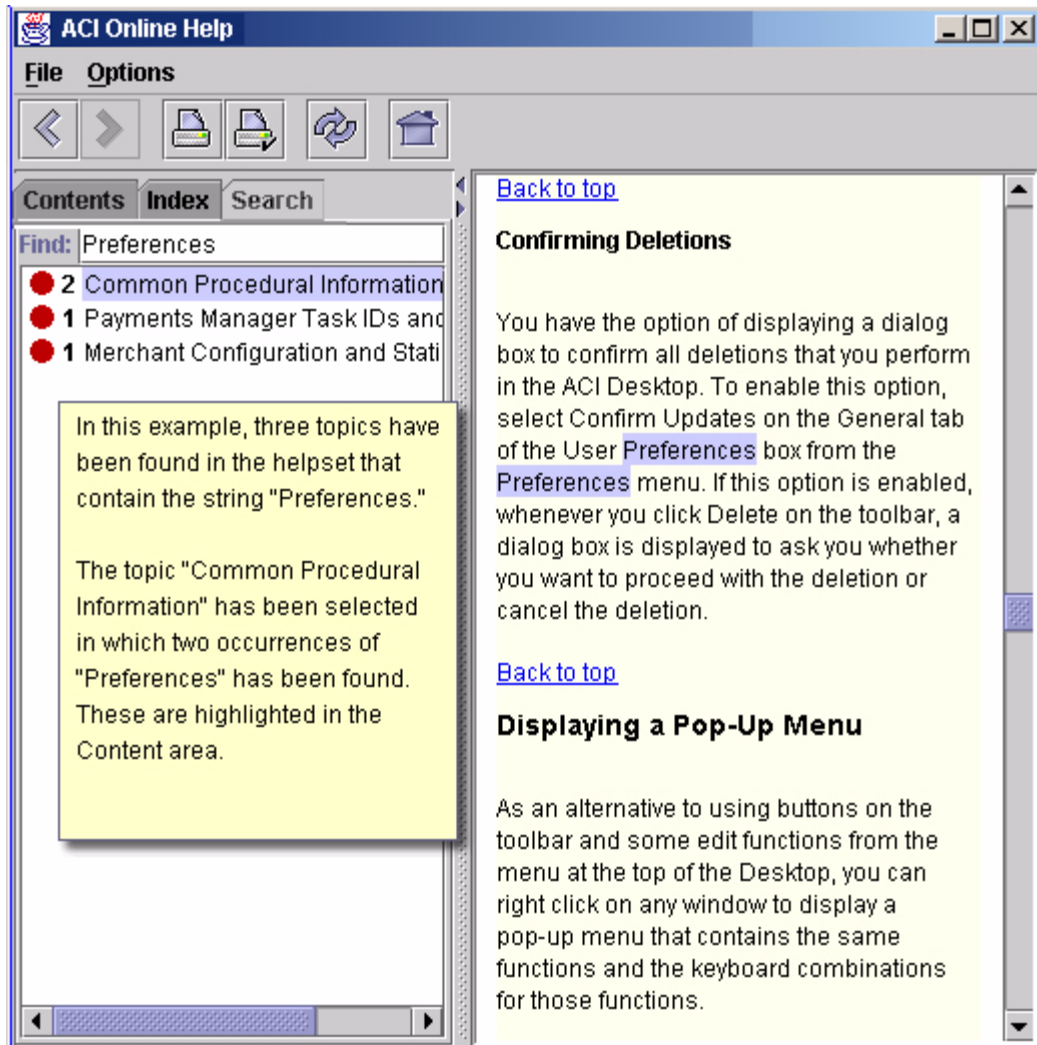


Search the helpset for a text string

The Search tab provides a text entry field (Find) to locate up to 100 occurrences of a text string within a helpset.

Note: Search is not case sensitive.

Enter the text string in the Find text entry field and press the **Enter** key. JavaHelp searches through the helpset and displays a list of topics in which the text string is found. This list is displayed in the Navigation Area by topic title and in descending order of number of matches. Select the desired topic to display the topic's content in the Content area.



Custom help content

Online help can be modified in a number of different ways depending on the level of modification desired.

- Content files can be modified.
- Cascading style sheets can be modified to change display formats.
- Images referenced in content files can be edited.

Note: Modifying the JavaHelp configuration files can adversely affect the performance of JavaHelp and is not discussed in this manual.

If an installed version of JavaHelp is customized, care must be taken to avoid overwriting the customized JavaHelp files when accepting an updated help JAR file for placement on the Version Checker repository. Therefore, the following guidelines can be used to establish procedures for maintaining and protecting customized JavaHelp:

- Always keep a backup of both original and customized files.
- Consider appending an identifier (e.g., “custom”) to the name of customized files. If a help file name is changed, the change must also be made to the map.jhm file. For example, a file named `f_connection.htm` file is customized and renamed to `f_connection_custom.htm`. The url attribute in the map.jhm file must be changed as follows:

From:

```
<mapID target="f_connection"
      url="f_connection.htm"/>
```

To:

```
<mapID target="f_connection"
      url="f_connection_custom.htm"/>
```

- Another option is to keep the same name of the file but place the customized file into a separate directory as follows:

```
<mapID target="f_connection"
      url="custom/f_connection.htm"/>
```

Customization of HTML content files

All content files within the JavaHelp JAR files are in compliance with the W3C HTML revision 3.2 format. This format is required for content files to display properly within the Helpset Viewer. Certain parts of the HTML revision 4.0 format can be used since it shares most markup tag elements with 3.2; however, care should be taken when implementing revision 4.0 features into content files destined for JavaHelp because certain revision 4.0 features might not be supported. To ensure proper behavior of content files in JavaHelp, always test the display characteristics of HTML files in JavaHelp before releasing the help files.

One advantage of the content files being in HTML format is that common editors can be used to modify help content. The specifics of editing HTML files are not discussed here since these depend on the editor; however, the following general always apply:

1. Before an HTML content file within a JAR file can be edited, the file must be extracted to a working location using a tool, such as WinZip. The extracted HTML file is a copy of the original file. Make a backup of the original file and place it in a specified archive directory for later retrieval. All editing should be done on the extracted copy.

2. Select an appropriate HTML editor. Note that some HTML editors automatically add embedded XML, HTML and other types of objects to HTML code. Such objects can adversely affect the display of the content within JavaHelp. As a result, editing tools that do not automatically add objects are recommended. Such tools include Macromedia's Dreamweaver and ASCII editors, such as Notepad.
3. Save the modified HTML file to the working location.
4. Add the modified file to the applicable JAR file replacing the original file in the JAR (the original being maintained in the archive directory location). Do not change the file extension.
5. If the name of the HTML file is changed to reflect customization, such as from `f_connection.htm` to `f_connection_custom.htm`, the URL attribute of the `map.jhm` file needs to be updated to reflect the name change. The `.jhm` file can be updated using an ASCII editor.

Change the display of content

The format and style of the content displayed in the Helpset Viewer can be modified by altering the values of the style properties in the applicable cascading style sheet (CSS) file. CSS files are defined in the `<head>...</head>` section of each HTML file used in the ACI online help using `<link>` tags as follows:

```
<link rel="StyleSheet" href="Templates/forScreen.css" type="text/css"
media="screen">

<link rel="StyleSheet" href="Templates/forPrint.css" type="text/css"
media="print">
```

CSS files are located in the Templates folder within the applicable helpset JAR (`ACI_Help_es.jar`, `ACI_Help_tss.jar`, and `ACI_Help_pm.jar`).

Note: For ACI Payments Manager, the name of the CSS is `document.css` for both screen and print media.

To ensure that content displays properly in the Helpset Viewer, always test before releasing the modified CSS files.

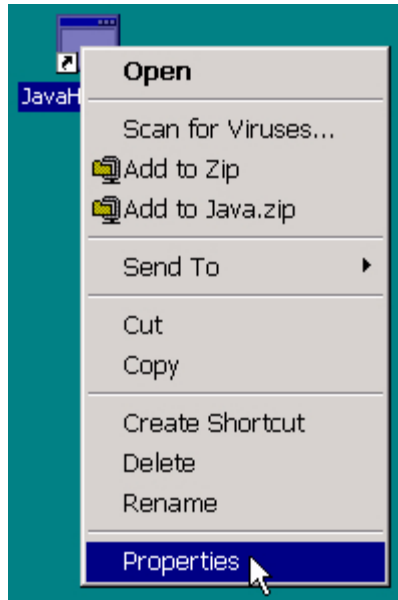
Install JavaHelp for stand-alone use

JavaHelp can be installed for direct access to helpset (`.hs`) files without the need to have the ACI desktop open. Note that Java must be already installed. The remainder of this topic discusses installing JavaHelp on a Java enabled Windows-based PC.

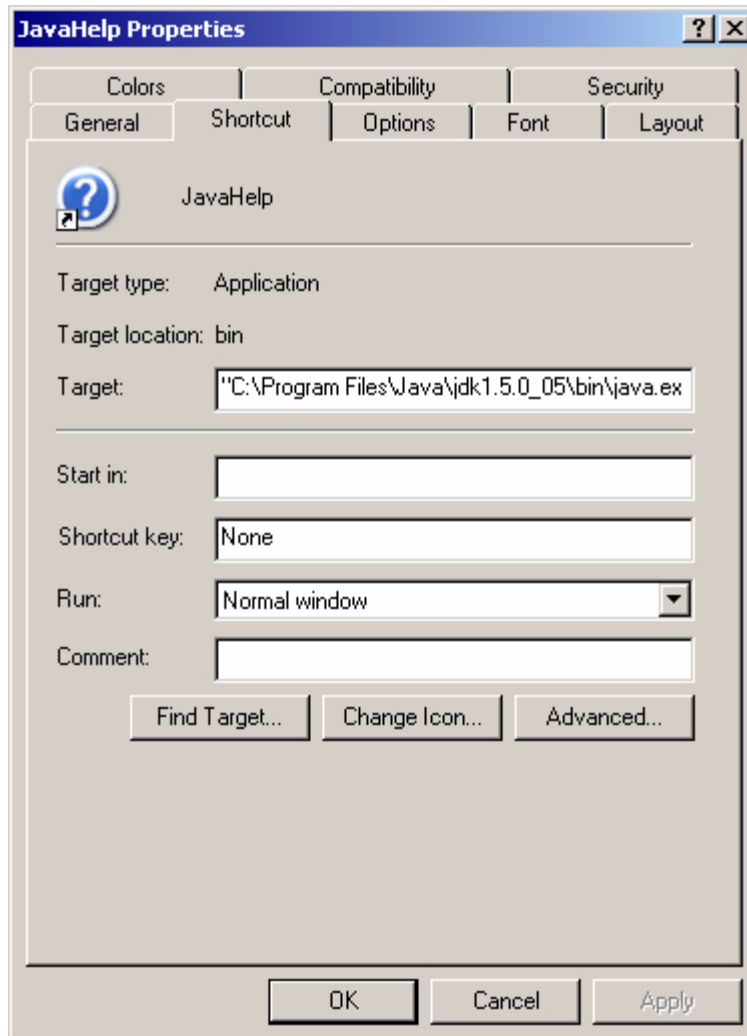
1. Download the latest version of JavaHelp from:
<http://java.sun.com/products/javahelp>

Note: JavaHelp is distributed as a zip file such as **Javahelp 2.0_02.zip** (a more current version might be available).

2. Open the zip file and extract all files to an easily accessible location, such as (e.g., C:\jh2.0). Among the files extracted is the **hsviewer.jar** file that is needed later in this procedure.
3. Find java.exe and create a shortcut to the file. Java.exe must be already installed. Cut and paste the shortcut to an easily accessible location.
4. Right click on the shortcut and select Properties on the drop-down menu:



The properties for the shortcut are available in the Properties window:



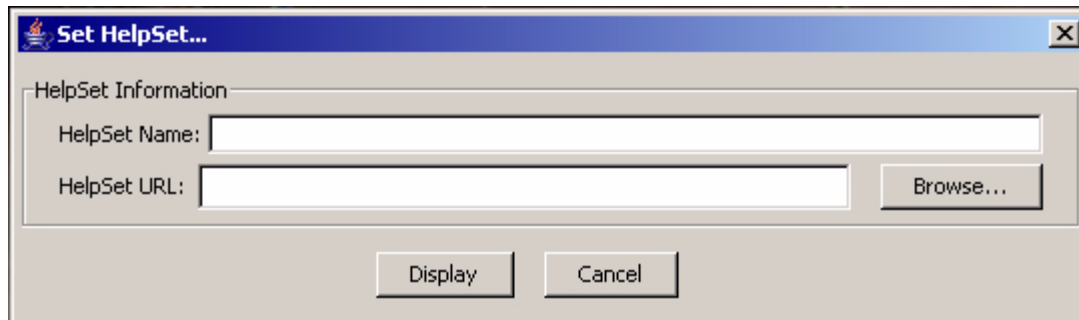
5. In the Target field located on the Shortcut tab, enter the path to java.exe, followed by a space, -jar, another space, and then the path to the installed location of hsvviewer.jar. In the illustration above, the full text of the Target field is partially obscured, but contains:

"C:\Program Files\jdk1.5.0_05\bin\java.exe" -jar "C:\jh2.0\demos\bin\hsvviewer.jar"

Note: If a blank space is located in a file path (e.g., C:\Program Files\...), enclose the path in quotes, as shown above.

6. Click **OK**.

7. Test the install by clicking on the shortcut, after which the following Set HelpSet window should display that can be used to browse to a helpset file (.hs extension) that you want to open.



If the above window does not display, a number of problems could exist. For example, either Java, JavaHelp, or both are not installed properly. Also, the target path in the shortcut could be incorrect. Verify that the paths in the shortcut reflect the actual paths for both `java.exe` and `hsvviewer.jar`.

8: BASE24-eps window (UI) information

Note: The window information previously provided in this section has moved to the *BASE24-eps Windows reference guide*.

9: Desktop.cfg file

The **desktop.cfg** configuration property file defines parameters for ACI Desktop user interface and Web clients. This file should be configured once for all users associated with a version checker server. Configuration files are created for each user of the desktop (*username.cfg*) and contain user-specific definitions. The properties of desktop.cfg file are divided into logical sections and are discussed below.

[Desktop] section

The **AppPath** property specifies the location of the UI application jars and must contain a valid directory location.

The **SSLEnabled** property is a boolean value indicating whether the desktop is to use an HTTPS connection for SSL support. This property only applies to those desktop connections to a webserver over HTTP.

The **CachePermissions** property is a boolean value indicating whether permissions are maintained in memory during user sessions.

The **EnableUSECReloadCacheCmd** property is a boolean value that if set to true enables those users who can access the User Security Role Configuration window to reload the user security authorization cache on demand by clicking the Reload Cache button. If this property is set to false, the Reload Cache button is not presented on the window.

The **Host** property specifies the name of the host group currently selected (last used) by the user. The host group specified by the value of the host property is used by the desktop to determine the connectivity parameters of the host server. The Host property is required.

The **HostTimeout** property specifies the amount of time, in seconds, that can elapse after a host request has been sent without a response being received. This property is optional and if specified must be a positive number. If the value is zero, or this property is missing, host requests waits 60 seconds for a response.

The **ShowLoggedInUser** property is used to activate the user icon located at the lower right of the desktop. When this icon is moused over, the desktop displays the name of the user currently logged onto the desktop in a popup message. Set this property to true to activate this functionality.

[Hosts] section

The [Hosts] section contains definitions for host groups used by the Host property above.

Each group (e.g., ProdServer) key entry within the [Hosts] section represents a logical host which consists of one or more physical hosts which are mirrors of each other and available for load balancing and fail over. The actual key name of each group entry is

listed in the host selection list on the desktop. The value of a group entry consists of one or more URL entries in the form of 'host:port/path' separated by commas ','. The currently selected group name key entry is set in the Host property of the [Desktop] section.

Example:

```
ProdServer=prod1:9023,prod2:8079
TestServer=test1:7765
```

[OnlineHelp] section

The **ACI Help** property identifies the default JavaHelp helpset (e.g., ACI_Help.hs) for the desktop.

[Common] section

The **Debug** property specifies whether or not debug messages are enabled for the desktop. Debug messages are sent to the debug log file identified by the **LogFileName** property located at the location specified by the **LogPath** property. Valid values are true or false (the default). This property can be specified at run time by setting the Debug system property on the command line. Setting the command line property overrides this property value.

The **TPA** property refers to Third Party Authentication. It specifies the type of third party authentication. If set to true then DEFAULT is implied. If set to false then NONE is implied.

The **ChangePasswordAllowed** property contains a boolean value that identifies whether the desktop allows password change capability. Typically used in conjunctions with the TPA property above. By default, this property is true. It can be set to false to disable the change password dialog of the desktop.

PasswordProperty defines the name and location of the SSL keystore encrypted password binary file.

The **ChannelTrace** is a boolean value. If true then connectivity and messaging between the desktop and server will be traced and logged to the log entry file configured for the desktop. If false, then no tracing will be logged. The default value, if not present, is false. If the system property -DChannelTrace is present it will be overridden by the setting from this property.

The **ChannelTraceLevel** property is a numerical value indicating the level of tracing that is performed. This value is only effective if the ChannelTrace property is set to true. Valid values are:

- 0 = Trace connection and all messages
- 4 = Trace all messages (default)
- 5 = Trace channel message header content only
- 6 = Trace summary information (this setting is only used until the user is authenticated after which the user's trace flag is used. See user preferences in section 2 of this manual)

[SSL] section

Refer to section 10 (SSL Keystore Password Manager) for more information.

Note: The SSL properties only apply if SSL enabled (i.e., the `SSLEnabled` property in the [Desktop] section is set to true).

If the **KeyStoreAutoUpdate** property is a boolean value that if set to true, the desktop automatically extracts the specified KeyStore file by identifying its location in the classpath. The extracted file will replace the local KeyStore file at the time this component is started. If this property is not present, it defaults to false. The KeyStore file name is found in the **KeyStore** property below.

The **KeyStore** property contains the key store file name used for retrieving private key information and any trusted certificates and associated keys. It must be built correctly in order for the SSL implementation to properly function. The default if not present is `defaultKS.sto`. Any specified file that is not present locally or if `KeyStoreAutoUpdate` is set to true, causes the component to search the classpath for the specified key store. If found, it is extracted and stored in the specified location (local).

The **KeyStorePassword** property specifies the password value that allows access to the KeyStore private and trusted certificate public keys. The default is "default".

The **KeyStoreType** property specifies the type of KeyStore file generated. The default, is JKS (Java Key Store). If a client implements a different type of keystore then it must be specified in this property.

The **TrustStore** property identifies the trust store where trusted certificates are accessed. By default, a separate trust store is not needed. This is an optional property and if not present, defaults to the value of the **KeyStore** property.

For the **TrustStoreType** property, refer to the `KeyStoreType` property above. If not present, it defaults to the `KeyStoreType` specified.

For the **TrustStorePassword** property, refer to the `KeyStorePassword` property above. If not present, it defaults to the `KeyStorePassword` specified.

The **Protocol** property defines the cryptographic protocol used, such as TLS (Transport Layer Protocol) or SSL (Secure Sockets Layer).

The **KeyMgmtAlgo** property specifies certificates standard and how keys are managed in regards to the trusted certificates. The default value, if not present, is "SunX509".

The **Provider** property specifies the Provider class that is used as the SSL implementation. The default value, if not present, is "com.sun.net.ssl.internal.ssl.Provider".

The **ProtocolHandlerPkgs** property specifies the protocol handler package that is required in conjunction with the provider implementation. The default value, if not present, is "com.sun.net.ssl.internal.www.protocol".

10: SSL Keystore Password Manager

The ACI desktop accesses SSL keystore passwords during secure communication with servers. These passwords are referenced (not stored) for the ACI desktop in the desktop.cfg configuration file. The keystore passwords are not stored in the desktop.cfg because passwords cannot be encrypted within the desktop.cfg file (see the PCI note below). Therefore, the encrypted passwords are placed in a binary file (e.g., passwords.bin) that is referenced within the desktop.cfg file in the [COMMON] section of file using the PasswordProperty property.

The Password Manager is provided to assist users in managing and encrypting SSL keystore password binary files. There are two types of passwords used: system and user-defined. The Password Manager can be used to manage both types. Passwords are associated with token IDs and descriptions.

Password values are encrypted using a proprietary bit encryption algorithm that the code base can cipher and decipher. The password properties file is made up of *tokens* (i.e., properties) that contain a password value. Thus, each password token is associated with a password value. It is the token IDs that are acted upon when using the Password Manager.



Payment Card Industry (PCI) Note: For security, stored password values must be encrypted. Cross-industry best practices recommend that vendor-supplied default passwords not be used, that passwords contain both numeric and alphabetic characters with a minimum password length of at least seven characters, that passwords be changed at least every 90 days, and that a new password not be the same as any of the last four passwords used.

Sample passwords.bin file

The example below is an ASCII-format representation of a passwords.bin file. The actual information is stored on disk encrypted and in binary format.

```
[UserTokens]
# This section is for user specific tokens (if any)

[SystemTokens]

ESF.password=USEC Database password token
EAF.password=UAUD Database password token
KeyStorePassword=SSL Key store password token for USEC/UAUD
TrustStorePassword=SSL Trust store password token for USEC/UAUD
```

```
[Password]
ESF.password=657e1f6fecea9ea8
EAF.password=657e1f6fecea9ea8
KeyStorePassword=512bb47f6f7206e90d0d440da775982b
TrustStorePassword=512bb47f6f7206e90d0d440da775982b
```

Run the Password Manager from the command prompt

When the Password Manager is initiated from a command prompt, both system and user-defined password tokens can be added and deleted.

To run the Password Manager from a command prompt, enter the following command syntax at the command prompt:

Syntax:

```
java -classpath {uif-jar} [-DPasswordProperty=password-bin-filename]
    framework.cipher.PswdMgr -keyword argument
```

where:

uif-jar is any one of the following JARs:

- *uif_dt.jar*
- *uif_sh.jar*
- *uif_verchk.jar*

password-bin-filename is the optional name of the password bin file entered as a system property. If it is not specified, it defaults to a value of "Password.bin". If the password bin file is entered as a run argument, it overrides this value.

keyword argument is one of the following formats:

```
-SetPW token password [password-bin-filename]
-VerifyPW token password [password-bin-filename]
-SetTKN token description [password-bin-filename]
-SetSYSTKN token description [password-bin-filename]
-RemoveTKN token password [password-bin-filename]
-RemoveSYSTKN token password [password-bin-filename]
-RemovePW token password [password-bin-filename]
```

where:

- ❑ SetPW — To set or change a password value and associate it to an existing token.
Note: A password cannot be changed unless the associated token for the password already exists. Therefore, use SetTKN first to add a user-defined or system password token.
- ❑ VerifyPW — To verify a password value.
- ❑ SetTKN — To add a user-defined password token.
- ❑ SetSYSTKN — To add a system password token.
- ❑ RemoveTKN — To remove a user-defined password token and the password associated with the token.
- ❑ RemoveSYSTKN — To remove a system password token and the password associated with the token.
- ❑ RemovePW — To remove a password value, but retains the token and description.

token is the name of a password property in the desktop.cfg (or local.cfg) file.

password is a clear password value. For the SetPW command function, you must enter the password twice to ensure it is entered correctly.

password-bin-filename is the optional name of the password bin file. If it is not specified, the value for the PasswordProperty system property is used.

description is a text description of a token. If the description contains spaces, it must be enclosed within quotes ("USEC database password key/token").

Run the Password Manager from the PswdMgrUI window

When running the Password Manager from the PswdMgrUI window, user-defined password tokens can be added and deleted. However, system password tokens cannot be added or deleted from the window, these tokens can only be changed.

To run Password Manager from the PswdMgrUI window, enter the following command syntax at the Command prompt. Tokens, descriptions, and password values can be edited using this window.

Syntax:

```
Java -classpath {framework-jar} framework.cipher.PswdMgrUI [password-bin-filename]
```

where:

framework-jar is any one of the following JARs:

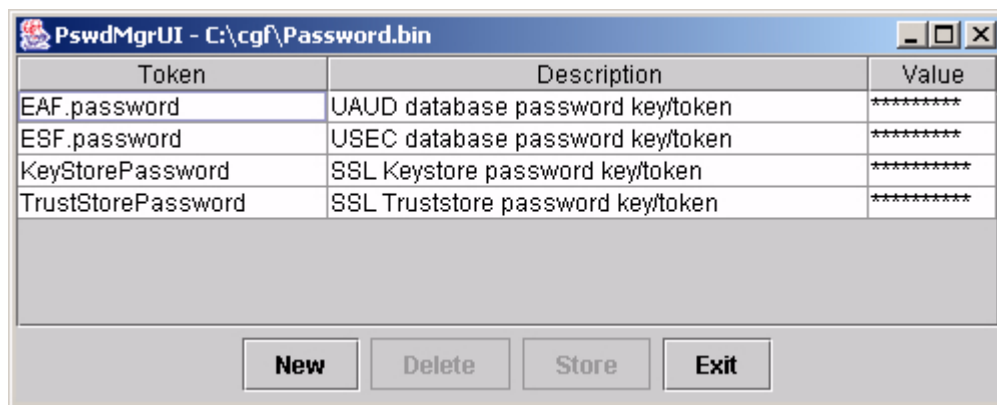
- uif_dt.jar
- uif_sh.jar

- uif_verchk.jar

password-bin-filename is the optional name of the password bin file. If it is not specified, it defaults to a value of "Password.bin".

PswdMgrUI window

The following is an example of the PswdMgrUI window. If the specified password bin file already exists, the window reads the file and displays all of the token/password entries in the file. System passwords are displayed in black text while user-defined passwords are displayed in blue text. In addition, all passwords values are masked with asterisk characters for security reasons. If the password bin file does not yet exist, no table rows are displayed.



Add a new user-defined password token

Use the following procedures to add a new user-defined password token.

1. Click the New button to insert a new empty row into the table.
2. In the empty row, enter a token ID, description, and password value.
3. Click the Store button to save the new information to the password binary file (e.g., password.bin).
4. Click the Exit button to stop the Password Manager.

Delete a user-defined password token

Use the following procedures to delete a password token.

1. Click on the row of the token that you want to delete. A selected row has a modified background color.
2. Click the Delete button to delete the token.
3. Click the Store button to save the change to the password binary file (e.g., password.bin).

4. Click the Exit button to stop the Password Manager.

11: Multibyte character data support

The BASE24-eps Multibyte Character Set Service module provides support for different alphabets, syllabic languages, and character sets that require two or more (multiple) bytes of data to represent a single character. This module provides support for additional languages and character sets independent of the number of bytes required to represent each character. The BASE24-eps Multibyte Character Set Service module enables BASE24-eps to store, display, and print user-entered data in local/native character sets such as Arabic, Han Chinese, Japanese, and Korean.

Configuration

You must have licensed the Multibyte Character Set Service module before you can configure its use.

Content encoding field

You configure use of the Multibyte Character Set Service module by setting the Content Encoding field on the Locale Content Configuration window (accessed from **Configure > Locale Content**) to a value of UTF-8.

Encoding properties

When you click Save on the Locale Content Configuration window, the value of the Content Encoding field is used to set both the MULTIBYTE_ENCODING Environment attribute in the Environment data source (Environment) and the MULTIBYTE_ENCODING Java property value in the Application Configuration data source (APPLCNFG). The MULTIBYTE_ENCODING Java property specifies how multibyte messages are encoded in the ACI desktop client and in the User Interface Framework (UIF) for messages containing multibyte fields.

Also note that the file.encoding Java system property must be set to a value of “ISO8859-1” when the ESWEB Java Server process is started (e.g., -Dfile.encoding=ISO8859-1) and to a value of “UTF-8” when an ACI desktop client is started (e.g., -Dfile.encoding=UTF-8 system property in the DesktopLauncher.lax file).

Because the MULTIBYTE_ENCODING Environment attribute and MULTIBYTE_ENCODING Java property must always be set to the same value, you cannot enter them on the Environment Configuration window or Application Configuration window. Refer to the **BASE24-eps Environment management user guide** for a detailed description of the MULTIBYTE_

ENCODING Environment attribute and the **BASE24-eps Java server reference guide** for a detailed description of the MULTIBYTE_ENCODING Java property and file.encoding Java system property.

Warmbooting the ESWEB Server process

After you save the value on the Locale Content Configuration window, you must either restart the ESWEB Server process or reload the configuration properties for the ESWEB Server process (e.g., P92^ESWEB) on the Configuration Properties window. You can access this window directly from the Locale Content Configuration window by clicking the “Reload Server Configuration Properties” link or you can manually navigate to the Configuration Properties window (accessed from **System Operations > Server Management > Server Administration > Configuration Properties**). To reload the configuration properties on the Configuration Properties window, enter the symbolic name of the ESWEB Server process in the Process ID field and click Reload.

Language preference

The locale you enter in the Language field on the General tab of the User Preferences window has no impact on multibyte character data fields. This setting only determines whether localized property files are used for the display of field labels, tooltips, screen messages, etc., on the ACI desktop. The values displayed in the selectable list for the Language field are controlled by the LocaleManager.language properties in the Desktop.properties file. Currently, only the U.S. English locale is configured for the BASE24-eps product.

Multibyte data fields

Once you have configured the ACI desktop to use a multibyte encoding value (e.g., UTF-8), multibyte fields are displayed on the windows, tabs, and fields listed in the following table. You can enter and display multibyte characters in these fields. These multibyte fields automatically determine whether text is displayed in a left to right or right to left direction based on the type of characters entered. Multibyte data fields are stored in multibyte data elements within existing data sources or in data sources that only contain multibyte data.

Window	Tab	Field	Data Source
Action Code Profile Configuration	Inbound Mapping	External Action Code Description (Locale)	Action Code External to Internal (ACT_CDE_EXT_INT)
	Outbound Mapping		Action Code Internal to External (ACT_CDE_EXT_INT)

Window	Tab	Field	Data Source
ATM Channel Configuration	Location (Locale)	All	ATM Static Data Table 1 (ATM_DS1)
	Migration (Locale)	All	
Card Management	General	Customer ID (Locale)	Card Account Multibyte (CARD_ACCT_MBYTE)
	Address (Locale)	Address (line 1 and 2) Postal Code	
	Accounts	Description (Locale)	
Institution Configuration	Address and Contacts (Locale)	All	Institution (INSTITUTION)
Merchant Channel Configuration	N/A	Merchant Name (Locale) (Display only)	Merchant Delivery Channel (CARD_ACCT_MBYTE)
	Terminal Information (Locale)	Terminal Location (Locale) Terminal Service Representative (Locale)	
Merchant Configuration	N/A	Merchant Name (Locale)	Merchant (MERCHANT)
	Contact Information (Locale)	All	

ATM consumer media and receipts

After the ACI desktop is configured to use a multibyte encoding, you can enter multibyte data on the ATM Consumer Media Configuration window, ATM Multi-byte Translation Configuration window, and on the ATM Receipt Format Profile Configuration window. This support is described in the ***BASE24-eps ATM configuration and management guide***.

Note: You do not actually enter multibyte characters on the ATM Multi-byte Translation Configuration window. The ATM Multibyte Translation data source (ATM_MBYTE_XLATE) maintained by this window is used to translate/map multibyte data to the corresponding single byte elements that the ATM is configured for in its language pages.

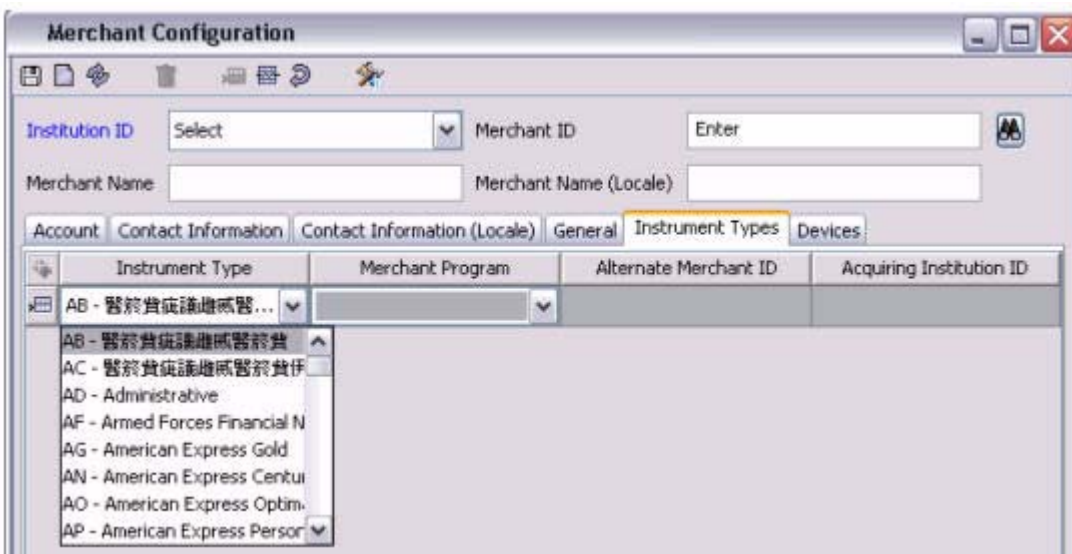
Instrument type descriptions

Multibyte instrument type descriptions are not supported on the Instrument Type Configuration window or in the Instrument Type data source (INSTRUMENT_TYPE). You can, however, override these descriptions with multibyte instrument type descriptions through the configuration of InstrumentTypeOverride properties in the CommonCodeValues property file. These override properties are in the format of InstrumentTypeOverride.<instrument-type-value>.text, where <instrument-type-value> is the two-character instrument type for which the description from the INSTRUMENT_TYPE is overridden.

When an InstrumentTypeOverride property is provided for an instrument type, then wherever that instrument type description is displayed on an ACI desktop window or browser page, it will contain the multibyte locale description from the property file rather than the ASCII description from the INSTRUMENT_TYPE.

The Instrument Type selectable list field on the Instrument Types tab of the Merchant Configuration window below depicts the display of multibyte instrument type descriptions using the localization override support entries in the CommonCodeValues property file. In the example shown, override entries are used for the AB and AC instrument types.

InstrumentTypeOverride.AB.text= 醫於費使議進威醫於費
InstrumentTypeOverride.AC.text= 醫於費使議進威醫於費伊



Scripts

After the ACI desktop is configured to use a multibyte encoding, you can add comments to scripts in a multibyte language and select the reading order of scripts (right to left instead of left to right) on the Script Editor window. You can also use certain exported operators in scripts to extract or report on multibyte data. This support is described in the **BASE24-eps Scripting manual**.

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JDOM Java XML API

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Web Services Description Language for Java Toolkit**

(wsdl4j-1.5.1)

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JavaBeans Activation Framework 1.2
Jython 2.2.1
MX4J 3.0.2 embedded Jetty implementation
SourceForge Piccolo XML Parser for Java
XML Security 1.2.1

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MX4J 3.0.2

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Duguay, Claude, "Visual Components: Swinging on a JCheckTree, JAVAPro Magazine, January 2001, Fawcette Technical Publications, Copyright 1999.

JSpinner

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